

Chapter 5

Carbon Pricing as a Key Policy for Climate Change Mitigation



5.1 Mounting Evidence in Support of Carbon Pricing

Carbon pricing is essential for achieving climate change mitigation in a cost-effective manner.

As discussed in Chapter 1, while developing Asia¹ has made significant progress, substantial decarbonization remains a major challenge as the region continues to pursue economic growth. It is therefore crucial for economies in the region to adopt emission reduction policies that minimize societal costs. By assigning a price to greenhouse gases (GHGs), carbon pricing creates incentives for businesses to shift from emissions-intensive products and processes to cleaner alternatives, spurring investment in low-carbon technologies and encouraging consumers to opt for low-emission goods and services. It helps align market behavior with climate objectives, making it a powerful and efficient mechanism for reducing GHG emissions and driving the transition to a low-carbon economy.

Robust evidence confirms that carbon pricing has been one of the most effective measures to reduce GHG emissions.

A major study of 17 established carbon pricing policies found that such systems reduced emissions by between 5% and 21% in their first few years of operation (Döbbeling-Hildebrandt et al. 2024). Similar results were also found through empirical analysis across 143 economies, 43 of which had a carbon price in place. The analysis showed that an additional price of €1 per metric ton of carbon dioxide (CO₂) helped reduce subsequent annual emissions growth by approximately 0.3 percentage points (Best, Burke, and Jotzo 2020). However, the performance of carbon pricing varies depending on the economy, sector, and policy environment being assessed (IPCC 2022).

There is a broad range of carbon pricing instruments available to governments. Subsidies for producing or consuming fossil fuels, as well as certain agricultural practices, can act as negative carbon pricing, which further encourages GHG-emitting activities. There are, however, a variety of carbon pricing instruments that discourage GHG emissions. The two primary instruments are carbon taxation and an emissions trading system (ETS), with both being cost-effective policy instruments for achieving environmental goals. Both can provide a clear carbon-price signal, either in terms of the cost (tax) of future emissions or the quantity of expected emissions reductions. They can also generate additional revenue for governments, which can then be used to support sustainable economic growth and “green” investment. International carbon markets, comprising both compliance and voluntary markets, can also help reduce emissions cost-effectively by leveraging the differences in abatement costs across economies.

Momentum is growing for the use of carbon pricing in the Asia and Pacific region.² Across the region, eight national carbon pricing initiatives had been implemented or were pending implementation at the time of publication. Japan and Singapore both employ carbon taxes, while Australia, the People’s Republic of China (PRC), Indonesia, Kazakhstan, the Republic of Korea (ROK), and New Zealand have all implemented a national ETS. Subnational ETSs are also employed by local jurisdictions in Japan. The Philippines, Thailand, and Viet Nam are considering adopting a domestic ETS, while India is developing a domestic national carbon market. Momentum is also growing for economies in the region to take advantage of international carbon markets, through both emerging compliance markets under Article 6 of the Paris Agreement on Climate Change and the voluntary carbon market.

The choice of instruments, and their combination, depends on a jurisdiction’s goals and circumstances.

This chapter discusses recent developments, best practices, and latest insights with regard to different carbon pricing policies, taking into account the regional context. To some extent, the design and implementation of carbon pricing policies are more important than the choice between the instruments.

¹ In this chapter, developing Asia refers to the 46 developing members of ADB.

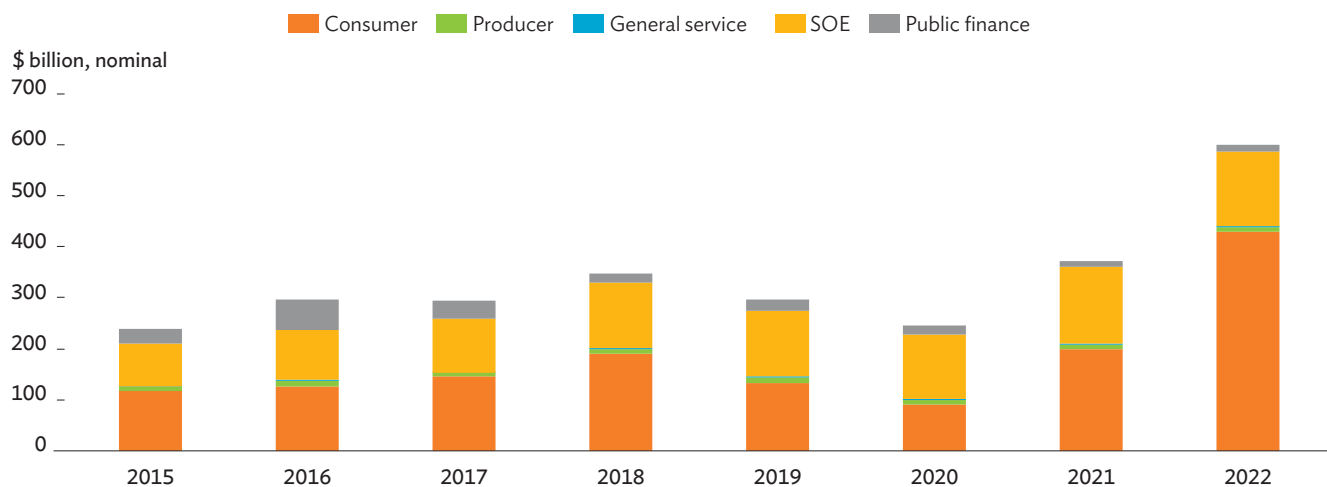
² In this chapter, Asia and the Pacific refers to the 46 developing members of ADB, along with Australia, Japan, and New Zealand.

5.2 Fossil Fuel Subsidies as Negative Carbon Pricing

5.2.1 High Public Financial Support for Fossil Fuels

In 2022, developing Asia provided \$600 billion in public financial support for fossil fuels. Acting as a negative carbon price and reinforcing dependence on fossil fuels, this support included \$440 billion in subsidies, at least \$146 billion in capital investments by state-owned enterprises (SOEs), and \$14 billion in international public finance (Figure 5.1). Together, this made up about a quarter of total global financial support to fossil fuels, which exceeded \$1.9 trillion in 2022.³ Public financial support for fossil fuels increases GHG emissions in two ways. First, subsidies for consumption directly reduce prices through measures such as price caps, direct transfers, and tax cuts, leading consumers to use more fossil fuels than they otherwise would. Second, support for new fossil fuel production capacity, for example by SOEs, incentivizes lock-in of new emissions-intensive infrastructure that must operate for many decades to recoup investments, creating “stranded asset” risks and slowing the transition to clean energy. A majority of fossil fuel subsidies in 2022 (\$429 billion) were directed toward consumption, particularly that of electricity (43%)⁴ and petroleum products (32%). Figure 5.2 shows that Kazakhstan, the Kyrgyz Republic, Tajikistan, Timor-Leste, and Uzbekistan dedicated a substantial share of national resources to fossil fuel subsidies in 2022.⁵ Production subsidies amounted to around \$11 billion, but this is likely to be an underestimate given many subsidies are undocumented or unqualified.

Figure 5.1: Public Financial Support for Fossil Fuels in Developing Asia, 2015–2022



SOE = state-owned enterprise.

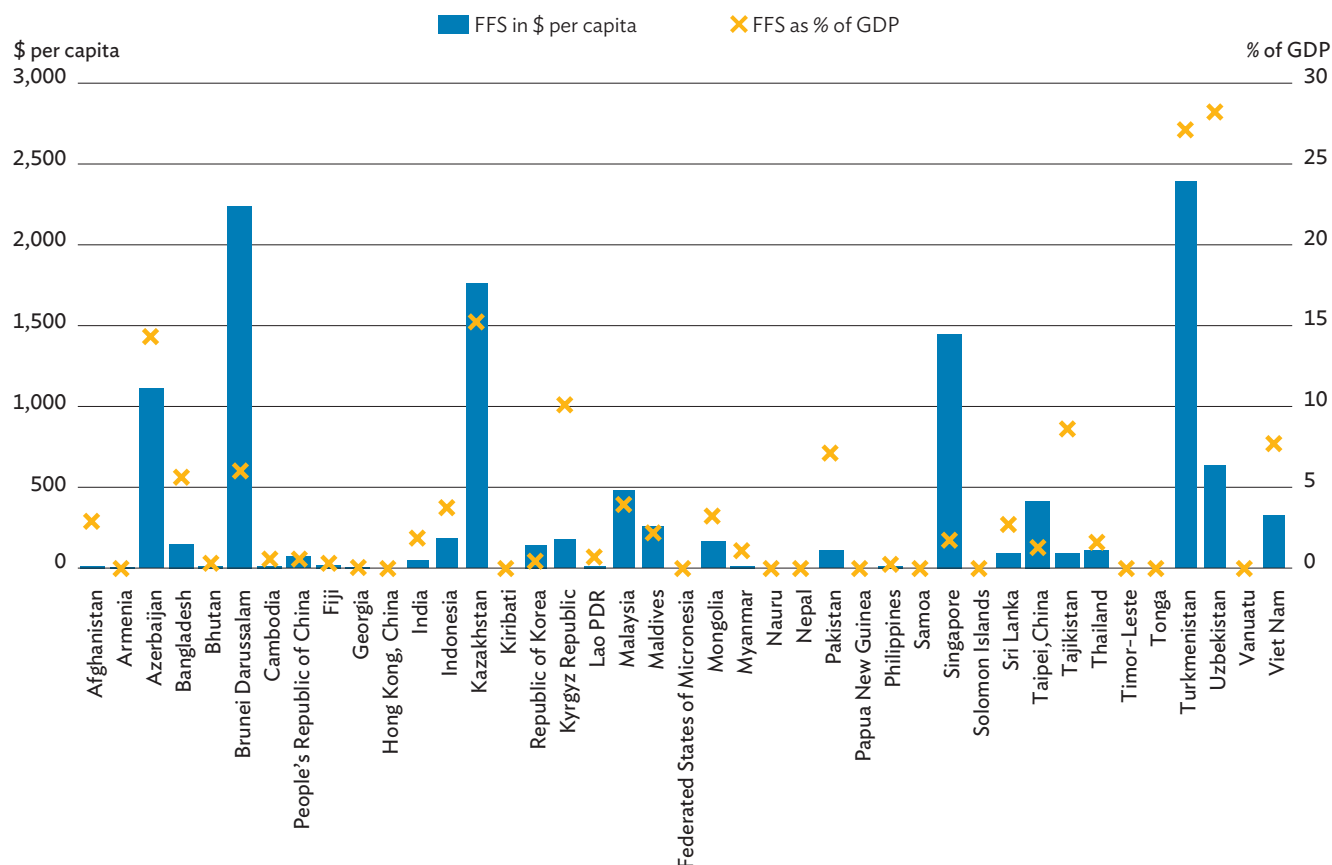
Sources: International Institute for Sustainable Development and Organisation for Economic Co-operation and Development. 2024. [Fossil Fuel Subsidy Tracker](#) (accessed 29 May 2024); Laan, T., A. Geddes, N. Do, L. Cameron, S. Goel, and N. Jones. 2023. Burning Billions: Record Public Money for Fossil Fuels Impeding Climate Action. *Energy Policy Tracker*; and Oil Change International 2024.

³ Subsidies of \$1.52 trillion (IISD and OECD 2024) plus capital investments by SOEs and lending from international public financial institutions, which totaled **\$0.35 trillion and \$0.02 trillion**, respectively, in G20 countries alone (Laan Geddes, Do, et al. 2023).

⁴ Electricity is considered a fossil fuel subsidy because the majority is generated from coal and natural gas.

⁵ Estimates of fossil fuel subsidies in this report include the use of the “price-gap” approach, which compares the end-use prices paid by fuel consumers with reference prices (such as import-parity prices). However, the ability to earn export or import-parity prices is often hindered by lack of sufficient export infrastructure, e.g., for natural gas and electricity from West and Central Asia.

Figure 5.2: Fossil Fuel Subsidies in Developing Asia, 2022



FFS = fossil fuel subsidy, GDP = gross domestic product, Lao PDR = Lao People's Democratic Republic.

Note: Population and GDP data were retrieved from the World Bank Data Repository (2023).

Source: International Institute for Sustainable Development and Organisation for Economic Co-operation and Development. 2024. [Fossil Fuel Subsidy Tracker](#) (accessed 29 May 2024).

SOEs, which dominate the supply of energy in developing Asia, are critical to the energy transition.

SOEs account for about 60% of energy investments in developing economies, compared to just 9% in developed economies. They especially dominate investment in fossil fuels and related networks in developing economies, accounting for 59% and 93% of total investment, respectively (IEA 2020). A review of governments in G20 economies found that, in 2022, SOEs invested \$348 billion in capital expenditure in coal, natural gas, and oil. The PRC, India, Indonesia, and the ROK accounted for over a third of this investment, totaling more than \$146 billion. However, actual SOE investment in fossil fuels is likely much higher due to the lack of a global dataset and poor transparency.

Continued investment in fossil fuel assets is a financial as well as a climate risk. Carbon pricing and “carbon border adjustment”⁶ are expected to increasingly impact demand for carbon-intensive products (ADB 2024). Investment in fossil fuels could therefore lead to stranded assets, which is a risk to taxpayers, given these are public assets. Conversely, SOEs can adopt carbon-reduction goals that bring huge investment

⁶ Carbon border adjustment is based on the premise that introducing carbon pricing in a jurisdiction provides an unfair advantage to imports from economies that are not subject to carbon prices. This could lead to higher demand for emissions-intensive imports. To prevent this carbon leakage, an adjustment is imposed on imports to ensure they are subject to a similar carbon price as domestically produced products.

opportunities to benefit both investors and the state (Fang and Jialu 2023). They can develop and implement evidence-based diversification strategies, and increase transparency about their investments, ambitions, and progress. This would be aligned with their institutional mandates, which typically go beyond the profit maximization of private sector counterparts, to include commitments to national energy security and social welfare. SOEs in the PRC, India, and Indonesia have responded to their governments' renewable and net zero ambitions, establishing road maps and investment packages for the deployment and integration of renewable energy sources (Fang and Jialu 2023; Tobing 2022; PLN 2021; Laan, Geddes, Bois von Kursk et al. 2023; Viswanathan et al. 2022).

Despite commitments to reform fossil fuel subsidies, progress has been slow. All developing economies in Asia and the Pacific have pledged to rationalize and phase out inefficient fossil fuel subsidies that encourage wasteful consumption through their participation in Asia-Pacific Economic Cooperation (APEC), the G7, the G20, or the United Nations (APEC 2021; G7 2022; G20 2021; UNDESA n.d.). However, as noted by van Asselt et al. (2023), only 29 of the 198 Parties to the Paris Agreement, including 12 from developing Asia, contain a reference to fossil fuel subsidy reform in their nationally determined contributions. Fossil fuel subsidy reforms have been slow, partly due to loopholes in current international commitments, including a lack of clear timelines.⁷ Before the energy crisis initiated by the Russian invasion of Ukraine, several developing Asian economies—notably Bangladesh, India, Indonesia, and Sri Lanka—had made progress on removing fossil fuel subsidies or increasing taxation. Box 5.1 explains relevant achievements in India and Indonesia, although some of this progress has been undermined by the reemergence of subsidies in both countries.

Fossil fuel subsidies tend to disproportionately benefit higher-income consumers. Fossil fuel subsidies for consumers are frequently justified as delivering universal energy access, being used as a form of social support, particularly in economies without alternative social welfare systems, and during periods of high energy prices. During the energy crisis in 2021–2022, many governments tried to shelter consumers from record-high fossil fuel prices, but only some of this support targeted vulnerable consumers (Sgaravatti et al. 2023; IEA 2023). Coady, Flamini, and Sears (2015) find that only \$1 of every \$10 in gasoline subsidies and \$1 of every \$30 in liquefied petroleum gas subsidies went to the bottom 40% of households in Asia and the Pacific. Evidence suggests that a more-effective policy to address cost-of-living pressures linked to energy pricing in the short term is through direct cash transfers to vulnerable households and targeted subsidies for critical energy services (Ari et al. 2022). Furthermore, examples in India and Indonesia show that a carefully designed and implemented phaseout of fossil fuel subsidies can result in the redirection of resources to social welfare, health, education, and clean infrastructure.

Governments can tackle subsidy reform by learning from successful international experiences. To act on fossil fuel subsidy reform, governments should develop national targets to align public resources with climate commitments, supported by detailed road maps tailored to local circumstances, evidence-based strategies, and firm deadlines for implementation. International experience in this area indicates that a reform plan should be based on inputs from appropriate government agencies and consultation with relevant stakeholders. A clear communication strategy to publicize the benefits of reform helps build political and public support. The appropriate pace of reform will be based on the type of subsidy, prevailing energy prices, and national circumstances. Finally, pricing mechanisms and SOEs need to be reformed to ensure subsidies do not reemerge (Beaton et al. 2013; Coady et al. 2013).

⁷ All current agreements refer to the phaseout of “inefficient” subsidies, but there is no universally endorsed definition of what makes a subsidy “inefficient” (OECD and IEA 2021). In addition, aside from the G7, which has committed to reform “inefficient” subsidies by 2025, all other commitments lack specific deadlines.

Box 5.1: Fossil Fuel Reform in India and Indonesia

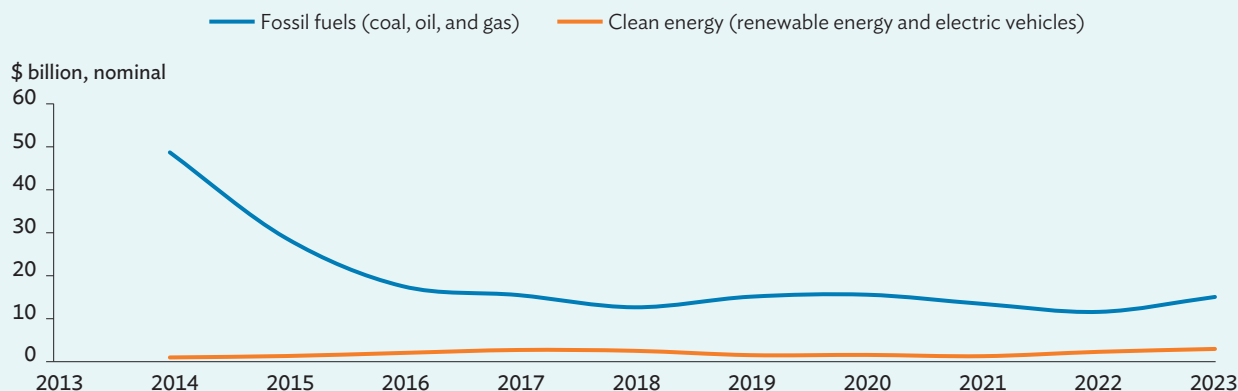
India: Shifting Support from Fossil Fuels to Clean Energy

Since about 2010, India has made noteworthy progress on fossil fuel subsidy reform through a calibrated “remove”, “target”, and “shift” approach (Bridle et al. 2019; Raizada et al. 2024). By carefully balancing the combined effect of three key policy levers—retail prices, tax rates, and subsidies on selected petroleum products—the country was able to reduce its fiscal subsidy in the oil and gas sector by 85%, from an unsustainable peak of \$25 billion in 2013 to \$3.5 billion in 2023 (Jain 2018; PPAC 2024; Raizada et al. 2024). India gradually phased out the subsidy on petrol and diesel (from 2010 to 2014) and carried out incremental tax increases (from 2010 to 2017), which created fiscal space to increase government support for renewable energy, electric vehicles, and strengthening of electricity infrastructure. The additional tax revenues from increases in excise duty on petrol and diesel from 2014 to 2017, a period of low international crude oil prices, were also redirected to improve access and target subsidies for expanding the use of liquefied petroleum gas (LPG) among the rural poor (Laan, Sharma, and Beaton 2018). Subsidies for LPG have since grown and may now require efforts to improve targeting and to incubate non-fossil-fuel cooking alternatives.

From 2010 to 2017, the Government of India introduced a cess (tax) on coal production and imports. Around 30% of the cess collections were channeled to a national clean energy and environment fund that supported clean energy projects and research (Government of India 2017). The cess significantly contributed to strengthening the budget of India’s Ministry of New and Renewable Energy during 2010–2017 and provided the initial funds for the country’s Green Energy Corridor scheme and its National Solar Mission, which helped bring down the cost of utility-scale solar energy and fund many off-grid renewable energy solutions. However, with the introduction of the goods and services tax (GST) in India after 2017, the cess on coal production and imports was subsumed within the country’s GST compensation cess, the flows of which were redirected to compensate states for revenue losses associated with the new tax regime (Garg et al. 2020).

As a result of India’s subsidy reforms and taxation measures, the country’s fossil fuel subsidies plummeted from 2014 to 2018, as shown in the figure below. Its renewable energy subsidies also reached a peak in 2017 but are now once again growing, with major support schemes targeting solar parks, state-owned enterprises (SOEs), and distributed renewable energy (Aggarwal et al. 2022).

India’s Subsidies for Fossil Fuels and Clean Energy, 2014–2023



Source: Raizada, S., D. Sharma, T. Laan, and S. Jain. 2024. *Mapping India’s Energy Policy 2023: A Decade in Action*. International Institute for Sustainable Development.

Indonesia: Shifting Support from Fossil Fuels to Social Protection

Declining oil production and mounting subsidy bills when fossil fuel prices spiked led the Government of Indonesia to implement major reforms in 2005, 2008, 2013, and 2014. In 2014, Indonesia implemented measures to reform fossil fuel subsidies, taking advantage of falling world oil prices. These measures included the removal of all gasoline subsidies except for distribution costs outside the Java-Madura-Bali area. Price caps on diesel were replaced with a fixed subsidy, allowing diesel prices to fluctuate according to international prices but remaining Rp1,000 per liter below the market price. This policy resulted in a saving for the government of roughly \$15.6 billion—over 10% of Indonesia’s annual expenditure in 2015.

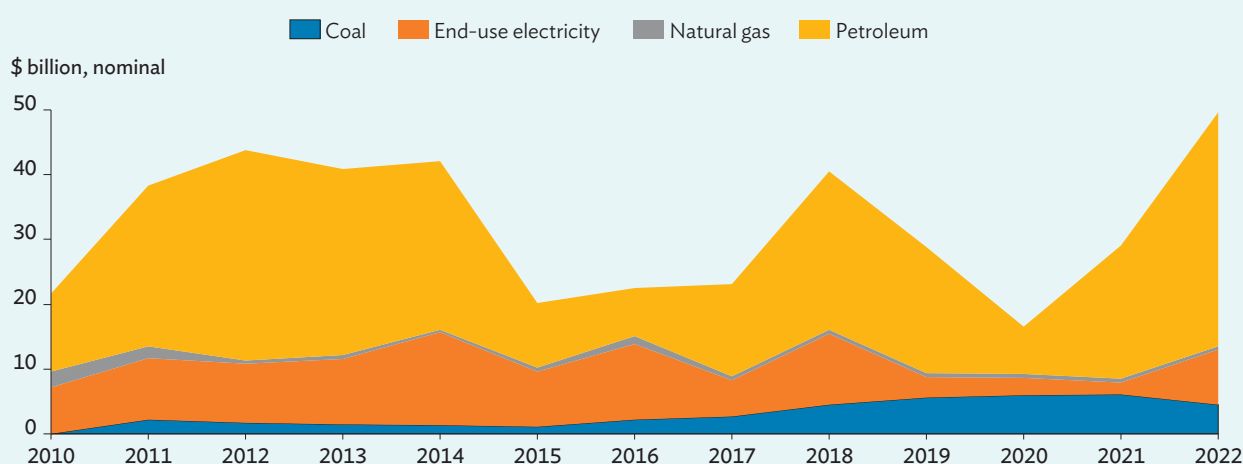
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Box 5.1 continued

A comparative analysis conducted by the International Institute for Sustainable Development demonstrates that the savings appear to have been reallocated to regional governments and villages, special programs in health insurance, housing for low-income groups, improved water access for all, and infrastructure investments by SOEs (Pradiptyo et al. 2016).

Despite Indonesia's efforts to reform its policies on fossil fuels, subsidy levels have surged again. From 2016 to 2020, the number of support measures for the Indonesian energy sector steadily increased, with a sharp rise in 2020, largely attributed to economic recovery packages that were mostly directed toward bailing out SOEs (EPT 2024). Most of Indonesia's support for fossil fuels comes from the government compensating the SOE Pertamina for the low retail prices it sets for gasoline, diesel, and some sales of LPG and kerosene (OECD 2023; Suharsono et al. 2022). This compensation rose sharply in response to the global energy crisis in 2021–2022, which induced a sharp increase in global prices of fossil fuels. As shown in the figure below, Indonesia spent around \$50 billion in 2022 to support fossil fuels, an increase of nearly \$35 billion on the amount spent in 2020 (IISD and OECD 2024).

Indonesia's Fossil Fuel Subsidies, 2010–2022



Source: International Institute for Sustainable Development and Organisation for Economic Co-operation and Development. 2024. [Fossil Fuel Subsidy Tracker](#) (accessed 29 May 2024).

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- Raizada, S., D. Sharma, T. Laan, and S. Jain. 2024. [Mapping India's Energy Policy 2023: A Decade in Action](#). IISD.
- Suharsono, A., M. Hendriwardani, T. B. Sumarno, J. Kuehl, M. Maulidia, and L. Sanchez. 2022. [Indonesia's Energy Support Measures: An Inventory of Incentives Impacting the Energy Transition](#). p. 51. IISD.

Source: Do, N., T. Laan, S. Raizada, L. Cameron, D. Sharma, O. B. von Kursk, and C. Beaton. Fossil Fuel Subsidy Reform. Background Paper for the *Asia-Pacific Climate Report 2024: Catalyzing Finance and Policy Solutions*. Asian Development Bank.

Reforms need to be complemented by policies to protect vulnerable consumers. Policy reforms around fuel subsidies may not be politically viable unless they also address a wide array of social goals. Removing subsidies can sometimes cause hardship for vulnerable populations and businesses and below-market fossil fuel prices are often seen as part of the social contract in many energy-producing economies (McCulloch 2023). Governments therefore need to identify vulnerable groups and businesses, anticipate impacts, and develop targeted assistance measures and compensation (such as cash transfers funded by subsidy savings) to address these impacts. In addition, some temporary financial support for the middle classes may also be needed to reduce the likelihood of political backlash. A commitment to reinvest subsidy savings in education, health, or cash transfers has been seen to double public support for subsidy reform (World Bank 2023a).

5.2.2. Agriculture Subsidies to Be Replaced by Climate-Smart Investments

Subsidies for agriculture and land use drive emissions and environmental degradation. After the energy sector, agriculture and land-use change combine as the second-largest source of GHG emissions, accounting for a quarter of all emissions worldwide. Globally, agriculture subsidies reached a historical high of \$851 billion per year for 2020–2022, a 2.5-fold increase from the levels of 2000–2002 (OECD 2023). The PRC and India accounted for 36% and 15%, respectively, of global agriculture subsidies in 2020–2022. About 74% of global agriculture subsidies go to producers in the form of market price support (such as import tariffs and other border measures) or payments for use of inputs (such as fertilizers and improved seeds). In parts of Asia, agriculture input subsidies have promoted such excessive overuse of fertilizers that yields have fallen and water pollution through increased nitrogen runoff has increased (World Bank 2023b). Similarly, subsidies that lower the price of electricity, diesel, and water for irrigation have promoted overuse and led to methane emissions, especially from rice cultivation (Rahut et al. 2023; Giardino, Wilson, and Hart 2023). Rice receives the highest share of subsidies among all commodities and is one of the top three sources of agriculture’s GHG emissions (OECD 2023). In some economies of developing Asia, state concessions in forest areas have effectively subsidized deforestation and increased both emissions and environmental degradation (ADB 2023a).

Agriculture subsidies are poorly targeted and exacerbate existing inequalities. Most agriculture subsidies accrue to wealthier farmers, even when programs are designed to target the poor (Ricker-Gilbert and Jayne 2016; World Bank 2023b). Consumer subsidy programs usually require significant out-of-pocket expenditure by recipients, which discourages participation from lower-income households. Meanwhile, producer subsidies tend to scale with levels of production, which means a disproportionate share goes to larger producers.

Reforming and repurposing agriculture subsidies can help farmers adapt to climate change. A majority of agriculture subsidies are concentrated in market-distorting measures such as market price support and use of inputs, while the share of subsidies that support research and development, extension services, innovation, and infrastructure is small and declining (OECD 2023). Investment in these areas has the potential not only to reduce emissions and environmental degradation but also to enhance the ability of farmers to adapt to the changing climate. Similar to the scenario with fossil fuels, agriculture subsidy reform should be based on a transparent process and multistakeholder consultation, with a clear communication strategy. It should also entail measures for those who face losses, such as cash transfers that target vulnerable people (OECD 2023; FAO, UNDP, and UNEP 2021).

5.3 Carbon Taxation as a Core Carbon Pricing Policy

In Asia and the Pacific in 2024, only Japan and Singapore had implemented a national carbon tax.

Japan began imposing a carbon tax in 2012, based on CO₂ emissions from the combustion of fossil fuels across all sectors. Japan's carbon price has been steady at ¥289 per metric ton of CO₂ equivalent (tCO₂e) since 2016. However, the Government of Japan provides exemptions and refund measures for specific fossil fuels used in the agriculture, energy, industry, and transport sectors. Singapore started imposing a carbon tax in 2019, targeting facilities in the energy, manufacturing, waste, and water sectors that directly emit at least 25,000 tCO₂e per year. Singapore has followed a phased approach to increasing carbon prices. Initially set at S\$5 per tCO₂e in 2019, the price increased to S\$25 in 2024, and is set to increase further to S\$45 by 2027 and to as much as S\$80 by 2030. In Japan and Singapore, revenues collected from carbon taxes will fund government decarbonization efforts, including investments in low-carbon technologies and practices and implementation of energy efficiency measures. In Singapore, revenues will also be used to cushion the impact of the carbon tax on businesses and households. From 2024, companies may use eligible international carbon credits to offset up to 5% of their taxable emissions. While carbon taxation is under consideration by more governments in the Asia and Pacific region,⁸ questions remain regarding policy design, macroeconomic implications, and impacts of a carbon tax on an economy's international competitiveness.

5.3.1 Carbon Tax Policy Design

Carbon taxation presents a compelling strategy for cost-effectively reducing GHG emissions. By letting polluters decide how to reduce their emissions, a carbon tax ensures that the reduction methods chosen are those that minimize costs, making it an economically efficient solution. Another primary argument for carbon taxation is its ability to raise substantial government revenue. In some cases, it can even be a more efficient revenue source than other types of taxation, making it a worthwhile policy consideration even if GHG emissions were not a concern. The revenue generated can be used to support public goods and services or to offset other taxes.

When designing a carbon tax, several elements must be considered. These elements include the tax rate, its coverage, use of the revenue, and the point of enforcement. The economic principle for setting the rate of a carbon tax is to align it with the harm caused by GHG emissions. Recent estimates suggest that the cost of the marginal damage from each extra metric ton of GHG emissions is around \$200 (Rennert et al. 2022), although there is considerable uncertainty and controversy over this figure. Prior to that, the best estimate was about \$40 per metric ton. This significant shift highlights the evolving understanding of how GHGs impact the economy.

Even low rates of carbon taxation can deliver significant environmental benefits. Despite the ideal of setting the carbon tax rate at \$200 per metric ton, such a rate may be politically challenging, especially in low- and middle-income economies. Therefore, alternative approaches could include setting a tax rate to meet specific emissions targets, such as those in the Paris Agreement, or coordinating tax rates with neighboring nations and major trading partners. The most realistic approach, however, is setting a tax rate based on what is politically feasible, because even relatively low tax rates can lead to substantial emissions reductions. For example, a modeling exercise shows that a carbon tax of \$12 to \$15 per metric ton could achieve emissions cuts similar to those expected from the Inflation Reduction Act in the United States (Bistline, Mehrotra, and Wolfram 2023).

⁸ Taipei, China is set to impose a carbon fee in 2024, which targets electricity and manufacturing facilities that emit more than 25,000 tCO₂e of GHG. The government has not set the fee rate but has announced that preferential pricing will be provided to entities with approved emissions reduction plan.

In principle, all GHG emissions should be taxed based on their relative warming potential. Process emissions, such as those from cement production, and emissions of non-CO₂ gases such as methane and nitrous oxide, are hard to monitor and measure. Land-use changes, such as deforestation, also present significant challenges. Given these difficulties, most economies have initially focused on energy-related CO₂ emissions. This focus poses dilemmas for economies where other sources are significant contributors of emissions. For example, in Indonesia, land-use change is a major emissions source that needs addressing despite the monitoring challenges.

Economic reasoning suggests that revenue from carbon taxation should be redirected to the highest-value public uses. Such uses might include offsetting distortionary taxes, compensating households through direct rebates or tax cuts, supporting affected industries, addressing trade and competitiveness issues, funding infrastructure development, or financing new environmental initiatives. The optimal use depends on specific context of an economy. In any case, the carbon tax revenue should not be treated differently from other public revenues just because it comes from protecting the environment.

The easiest method for enforcing a carbon tax is to tax fossil fuels upstream. This method involves taxing coal, oil, and natural gas at natural “choke points” in the supply chain, such as import points or refineries. This approach simplifies enforcement by focusing on a small number of entities, minimizing opportunities for tax evasion and avoidance. It has considerable advantages for lower- and middle-income economies in particular, as these economies usually have large informal sectors and limited enforcement capacity. However, this approach only works for energy-related CO₂. Other emissions, such as process emissions and non-CO₂ gases, must be enforced downstream, where the actual emissions occur, which complicates enforcement.

5.3.2 Challenges and Design Responses of Carbon Tax Policy

Implementing a carbon tax policy presents several significant challenges. The most prominent of these challenges are concerns around distributional impacts, uncertainty on emissions reductions, and international trade and competitiveness (covered later in this chapter). These issues need careful consideration and strategic responses to ensure the policy’s effectiveness and fairness.

A carbon tax may have regressive effects, potentially harming low-income households. This scenario is largely due to the increased costs of energy goods, which are often necessities. However, the regressive nature of a carbon tax may be less severe than initially perceived. Many of the effects of a carbon tax occur through the prices of other goods, some of which are much less regressive. For instance, higher prices for air travel primarily impact wealthier consumers. Moreover, even the energy price effects may be less regressive in developing economies than in the developed economies because what is considered a necessity, such as cars in the United States, might be viewed as a luxury in lower- and middle-income jurisdictions. Meanwhile, the income effects of a carbon tax might be progressive. Evidence from high-income economies indicates that a carbon tax tends to reduce capital income more than labor income, and capital income is mainly accrued by higher-income households. This progressive income effect can offset some of the regressive impacts on consumption.

The potentially regressive impacts of a carbon tax can be balanced by appropriate use of revenue. The first option under such a strategy is through per capita rebates that distribute the revenue generated from the carbon tax equally among all citizens. Another option is income-targeted transfers, i.e., direct financial assistance to low-income households to offset increases in energy costs. Tax revenues can also be used to fund anti-poverty initiatives that support low-income communities. Last, reducing other regressive taxes can also help alleviate the financial burden on vulnerable households. Concerns about specific industries or firms can be addressed through output-based rebating or marginal exemptions. These measures might reduce policy efficiency slightly, but they can effectively mitigate undesirable distributional effects at a relatively low cost.

Imposing a carbon tax may not provide certainty on emissions reductions. Carbon taxes provide price certainty, but they leave emissions levels variable. If emissions certainty is a major concern, an adjustment mechanism can be introduced whereby tax rates are adjusted according to a target emissions path. If emissions are above target, the carbon tax rate is increased and vice versa. Although this approach sacrifices some price certainty, it helps stabilize emissions by creating a hybrid policy between a fixed carbon tax and an emissions trading system (ETS).

5.3.3 Macroeconomic Implications of a Carbon Tax

It has been mooted that a carbon tax could hinder economic growth. When considering adoption of a carbon tax, a key question is the potential impact on the broader economy, particularly on economic growth, inflation, and employment. Understanding these macroeconomic implications is crucial for designing an effective and sustainable carbon tax policy.

Economic analysis suggests the economic impacts of a carbon tax are moderate. Model-based simulations indicate that the impact of a carbon tax on economic growth is likely to be small and could even be slightly positive (Goulder and Hafstead 2013; Lu, Tong, and Liu 2010; McKibbin et al. 2015). The effect tends to be more positive if the carbon tax is applied upstream, meaning at the point of fossil fuel extraction or import, and if the revenue generated is used to reduce other inefficient taxes. The primary impact of a carbon tax is not to reduce growth but to shift the economy from high-carbon to low-carbon activities, thereby affecting the carbon intensity of the economy rather than overall gross domestic product. Empirical research on existing carbon pricing strategies shows that established mechanisms have not significantly impacted economic growth rates once implemented (Köppl and Schratzenstaller 2023). Studies consistently find no large positive or negative effects on economic growth, with confidence intervals centered around zero, indicating minimal impact (Bernard, Kichian, and Islam 2018; Metcalf and Stock 2020). However, it should be noted that results may change if evaluations are done over longer time frames, focus on high rates of carbon taxation, and/or take into account international interactions. For instance, modeling of global economic impacts up to 2050 shows that boosting green investment using a combination of climate mitigation policies, including carbon pricing, can increase global output for the first 15 years, but output reverts below baseline levels once the impacts associated with stimulating investments wear off (Jaumotte, Liu, and McKibbin 2021).

Another concern is the potential for a carbon tax to drive inflation. However, evidence suggests that there are no significant inflationary effects following the implementation of a carbon tax. This finding is somewhat surprising, as economic models typically suggest that a carbon tax would pass through to consumer prices, potentially raising overall price levels. Research by Konrad and Weder di Mauro (2023) indicates that potential price increases are either offset by other factors or do not materialize as expected, leading to no significant change in inflation. The study suggests that proper redistribution of revenue and appropriately calibrated central bank interventions can neutralize any inflationary pressure from carbon pricing policies.

A key issue is that a carbon tax can impact employment, particularly in energy-intensive industries.

While there may be job losses in carbon-intensive industries, these are often offset by employment gains in low-emissions sectors, resulting in a net shift rather than a net loss of jobs (Yamazaki 2017; Shapiro and Metcalf 2023). Nevertheless, this job shift can pose challenges for workers in affected industries. To mitigate these effects, it is advisable to preannounce or phase in the carbon tax, allowing more time for the labor market to adjust. Additionally, implementing policies that provide compensation or retraining for workers in impacted sectors can help ease the transition.

5.3.4 Wider Adoption of Carbon Taxation

It is vital that politicians and the general public be united in their climate ambitions. Perhaps the most formidable challenge facing the widespread adoption of a carbon tax is securing sufficient political will and gaining public acceptance. Unlike technical feasibility, political and societal readiness can be unpredictable. These challenges can be navigated through strategically planned measures such as rebates to vulnerable households, transition assistance for affected industries, clear and transparent preannouncements on policy, and gradually phasing in the tax regime. However, despite these efforts, significant resistance from various stakeholders can remain, which is when implementing an ETS becomes a viable alternative for many economies, as discussed in Box 5.2.

Box 5.2: Carbon Taxation Versus Emissions Trading System

A carbon tax and an emissions trading system (ETS) are two main carbon pricing policy instruments. In an ideal world, they are equivalent in efficiently achieving environmental objectives. However, in practice, there are some notable differences between the two policies.

The first major advantage of carbon taxation over an ETS is the price certainty. A carbon tax provides a clear and predictable price path for greenhouse gas (GHG) emissions. This predictability is crucial for businesses as they plan long-term investments. Knowing the future cost of GHG emissions allows companies to make informed decisions without the risk of facing volatile prices that could drastically change in a short period. This stability is especially valuable for capital-intensive industries where investment horizons span several years.

Another key benefit of a carbon tax is its relative simplicity and transparency. Unlike an ETS—which requires complex trading mechanisms, continuous cap-setting, allocation mechanisms, and monitoring of trades—a carbon tax is straightforward to implement. It clearly outlines the costs associated with GHG emissions, making the financial flows easy to track and understand. This transparency simplifies the process for both the government and businesses, reducing administrative burdens and enhancing public trust in the system.

A carbon tax can also generate substantial revenue for the government and this revenue can be used to fund various public services or offset other taxes. While ETSs can also generate revenue if permits are auctioned, this is not inherently built into the system: in reality, most emissions permits are given to emitters for free.

Furthermore, the implementation of a carbon tax via the relevant tax authority offers a distinct advantage in lower- and middle-income economies. By leveraging the existing infrastructure and expertise of tax authorities, a carbon tax can be more effectively enforced, ensuring better compliance and outcomes. By contrast, environmental ministries in these economies may lack the necessary capacity and resources to build and regulate an ETS.

However, ETSs do have several notable advantages over carbon taxation, particularly in terms of emissions certainty. An ETS provides a clear limit on emissions by setting a cap, allowing the market to determine the price of emissions permits. This approach ensures that total emissions remain within a predefined limit, which can be set to meet specific environmental targets.

Distributional flexibility is another significant advantage of an ETS. The allocation of permits can be adjusted to support particular groups or industries, allowing targeted economic support while still adhering to overall emissions limits. While similar effects can be achieved with tax rebates in a carbon taxation system, the inherent design of an ETS makes such adjustments more straightforward. Additionally, the allocation of permits can be done discreetly, without drawing public attention to the redistribution of resources. This can be advantageous in situations where explicit financial transfers might be politically sensitive or controversial.

However, broadly speaking, these two carbon pricing policies are more similar than they are different, with both approaches sharing the common goal of reducing GHG emissions and mitigating climate change at the lowest economic cost. Design and implementation features probably matter more than the carbon taxation versus ETS debate.

Source: Authors.

5.4 Build and Connect Emissions Trading Systems

5.4.1. Emissions Trading System Practices in Asia and the Pacific

Of the 13 national ETSs established worldwide by 2024, six are in the Asia and Pacific region.⁹ National ETSs of various types operate in Australia, the PRC, Indonesia, Kazakhstan, the ROK, and New Zealand. New Zealand's ETS dates back to 2008 and is the second oldest in the world.¹⁰ It was followed by Kazakhstan and the ROK in 2013 and 2015, respectively. By 2016, the PRC had initiated pilot ETSs in eight cities¹¹ and established a nationwide ETS in 2021. Australia does not have a formal ETS, but has operated a mandatory baseline-and-credit system since 2023.¹² Indonesia's ETS commenced in 2023.

The region's six national ETSs differ in terms of their ambitions and implementation. All of them, however, are central to meeting the GHG mitigation commitments of their respective countries: to achieve net zero by 2050 for Australia, the ROK, and New Zealand; and by 2060 for the PRC, Indonesia, and Kazakhstan (Table 5.1). In addition, all of these economies are also aiming for significant GHG reductions by 2030. This is noteworthy especially for the PRC, Indonesia, Japan, the ROK, and Australia, which are the world's first, sixth, seventh, 13th, and 15th largest emitters of GHGs, respectively (European Commission 2024). In terms of scope and coverage, the ROK's ETS is the most comprehensive, as it includes all of the GHGs nominated in the Kyoto Protocol. New Zealand's ETS is also comprehensive, though it excludes biogenic methane, which is a major emissions category for that jurisdiction. Most of the other national ETSs focus primarily on CO₂ emissions. The share of total emissions covered by ETSs ranges from a low of 26%–29% (Australia and Indonesia) to a high of 88.5% (ROK). The ROK and New Zealand have the most comprehensive coverage in terms of sectors, while the other systems mostly focus on the energy sector, a major source of CO₂ emissions.

There is very little auctioning of emissions allowances in any ETS in Asia and the Pacific. Allowances within the region's ETSs are mostly given for free, especially for emissions-intensive and trade-exposed industries. In New Zealand's ETS, 54% of allowances were offered for auction in 2022, but none was sold, since the clearing price was lower than the reserve price (ICAP 2022). With regard to allowance markets, only two ETSs (the ROK and New Zealand) had a long enough time series for prices.¹³ Allowance prices have been low in the ROK and moderate in New Zealand, especially as compared to the ETS of the European Union (EU) as shown in Figure 5.3. Additionally, in line with the experience of the EU's ETS, allowance prices in the ROK and New Zealand have shown volatility. In the latter, prices fell to practically zero (\$2) in 2013—soon after the announcement delinking the New Zealand ETS from Kyoto markets—but prices have been mostly rising since then.

⁹ The 13 national ETSs worldwide are found in Australia, Austria, Canada, the PRC, Germany, Indonesia, Kazakhstan, the ROK, Mexico, Montenegro, New Zealand, Switzerland, and the United Kingdom.

¹⁰ The first ETS for GHGs was the European Union's ETS in 2005. Box 5.3 provides a brief overview of the EU's ETS.

¹¹ The eight cities are Beijing, Chongqing, Fujian, Guangdong, Hubei, Shanghai, Shenzhen, and Tianjin.

¹² Unlike an ETS, in a baseline-and-credit system there are no emission caps. Firms below a given baseline can generate and sell credits to other firms that exceed the baseline.

¹³ The PRC's ETS market is nascent, but data for 2 years are shown in Table 5.1. Kazakhstan's ETS had stops and starts and allowance prices are not significantly different from zero (Table 5.1, row 10).

Table 5.1: Main Features of National Emissions Trading Systems in Asia and the Pacific

	Australia	PRC	Indonesia	Kazakhstan	ROK	New Zealand
1. GHG reduction targets for the country	By 2030: 43% below 2005 levels By 2050: Net zero emissions	By 2030: 65% below 2005 levels By 2060: Carbon neutrality	By 2030: 31.9% below BAU By 2060: Climate neutrality	By 2050: 40% reduction in energy sector emissions from 2012 levels. By 2060: Carbon neutrality	By 2030: At least a 35% reduction below 2018 emissions By 2050: Carbon neutrality	By 2030: 50% reduction of net emissions below gross 2005 levels By 2050: Reduce net emissions of all GHGs (except biogenic methane) to zero
2. Year ETS introduced	2016	2021	2023	2013	2015	2008
3. GHGs covered	CO ₂ , CH ₄ , N ₂ O, SF ₆ , HFCs, PFCs	CO ₂	CO ₂ , CH ₄ , N ₂ O	CO ₂	CO ₂ , CH ₄ , N ₂ O, SF ₆ , HFCs, PFCs	CO ₂ , CH ₄ , N ₂ O, SF ₆ , HFCs, PFCs (but no biogenic CH ₄)
4. Covered emissions (% total emissions and year)	26% (2020)	40% (2021)	29% (2023)	47% (2021)	88.5% (2021)	48% (2021)
5. Emissions subject to ETS	137 MtCO ₂ e	5,200 MtCO ₂ e (approx.)		160 MtCO ₂ e	599 MtCO ₂ e	35.9 MtCO ₂ e
6. Sectoral coverage	Waste, Domestic Aviation, Transport, Industry	Power	Power	Industry, Energy	Waste, Domestic Aviation, Transport, Buildings, Industry, Energy	Forestry, Maritime, Waste, Domestic Aviation, Transport, Buildings, Industry and Energy
7. No. of obligated entities/firms (year)	215 (2023)	2257 (2021–2022)	63 entities, 146 installations (2024)	128 companies (201 installations)	804 (2023)	4114 (2023)
8. Allowance allocation	Set up on the basis of production levels multiplied by an emission intensity value	Output based benchmarking for 4 benchmarks: conventional coal plants below 300 MW; above 300 MW; unconventional coal and natural gas. Free allocation currently.	Auctioning, and benchmarking	Free allocation (benchmarking and grandparenting approach)	Free allocation (benchmarking and grandparenting approach) and auctioning	Free allowance and auction Note: Proportion of allowances offered at auction in 2023 was 54% but none were sold (winning bid less than reserve price).
9. Compliance period	Annual	Annual	Annual	Annual	Annual	Annual
10. Average prices (2023)		Secondary market: CNY68.35 (\$9.65)	Auction: Rp10,000 (\$0.64) Secondary market: Rp69,600 (\$4.45)	Secondary Market: T473 (\$1.04)	Auction: W10,672 (\$8.17) Secondary market: W9,999 (\$7.66)	Auction: NZ\$0 (\$0) Secondary market: NZ\$62.79 (\$38.30)

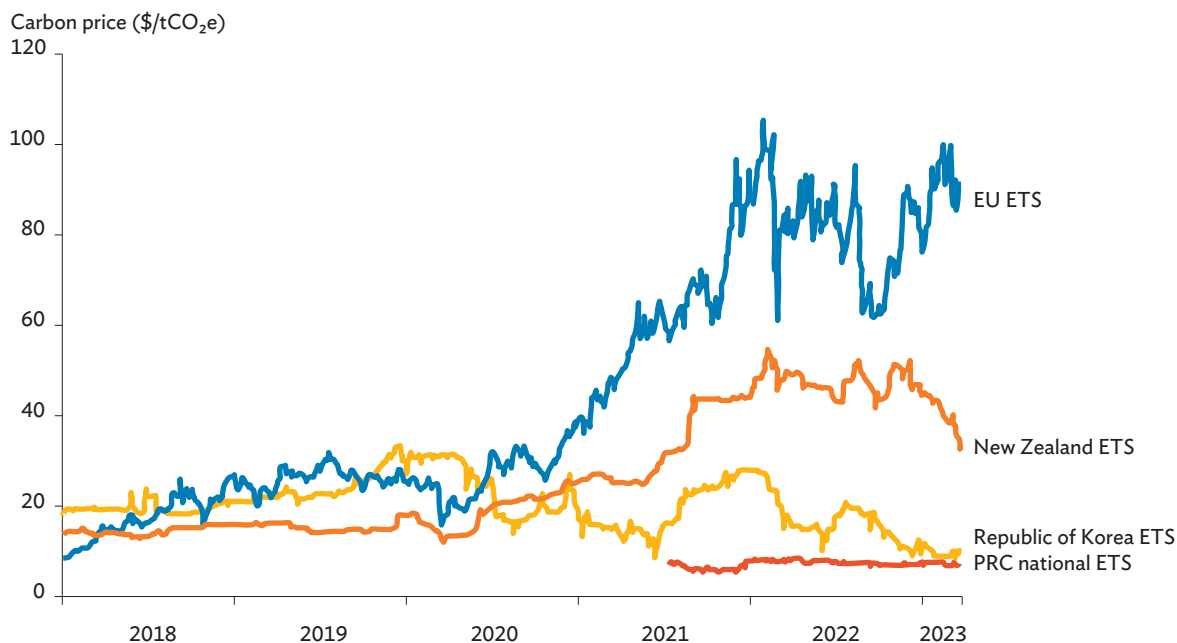
continued on next page

Table 5.1 continued

	Australia	PRC	Indonesia	Kazakhstan	ROK	New Zealand
11. Monitoring, reporting, and verification	Monitoring: Facilities need to apply for a site-specific emissions intensity value for the first compliance year. Reporting: Annual self-reporting according to the guidelines defined in the “National Greenhouse and Energy Reporting Act.” Verification: All facilities with emissions greater than 1 MtCO₂e per year are required to undergo independent emissions audit each year.	Monitoring: Covered entities are required to set up monitor plans and monitor their emission based on these plans. Reporting: Covered entities must submit the previous year's emissions reports by the end of April each year. Verification: Provincial-level ecological and environmental authorities are responsible for organizing the verification of GHG reports. Verification of emissions from the energy sector must be complete by the end of June.	Monitoring: An MRV system is currently in operation in the industry sector and the power generation sub-sector. Pilot MRV programs are being conducted also in the cement and fertilizer sectors. Reporting: The reports must be submitted by the end of January for the preceding year. Installations must report emissions of considered GHGs expressed in units of CO₂e. Verification: It should be completed by the end of March, following the January reporting deadline. Verifiers are required to adhere to the guidelines for GHG emission verification in the power subsector.	Reporting: Required annually for installations above the 20,000 MtCO₂e/year threshold. Verification: Emissions data reports and their underlying data require third-party verification by an accredited auditor.	Reporting: Annual reporting of emissions from the previous year must be submitted by the end of March. Verification: Emissions must be verified by a third-party verifier. Framework: Emission reports are reviewed and certified by the Certification Committee of the Ministry of Environment by the end of May.	Reporting: Most sectors are required to report annually; the deadline is the end of March to submit an annual emission return. Verification: follows a system of self-reporting supplemented by a program of official government audits. Third-party verification is not required for emissions reports. However, participants must seek third-party verification if they apply for the use of a unique emissions factor, as opposed to using the default factors supplied by the government.

BAU = business as usual; PRC = People's Republic of China; CH₄ = methane; CO₂ = carbon dioxide; CO₂e = carbon dioxide equivalent, ETS = emissions trading system; GHG = greenhouse gas; HFC = hydrofluorocarbon; MRV = monitoring, reporting, and verification; MtCO₂e = metric tons of carbon dioxide equivalent; MW = megawatt; N₂O = nitrous oxide; PFC = perfluorocarbon; ROK = Republic of Korea; SF₆ = sulfur hexafluoride.

Source: Gupta, S., S. Basu-Das, and S. Gupta. 2024. Carbon Pricing Through Emissions Trading In Asia-Pacific: A Comparative Analysis and the Way Forward. Background paper for the *Asia-Pacific Climate Report 2024: Catalyzing Finance and Policy Solutions*. Asian Development Bank.

Figure 5.3: Comparison of Carbon Prices in Selected Economies in Asia and the Pacific

PRC = People's Republic of China, ETS = emissions trading system, EU = European Union, tCO₂e = metric ton of carbon dioxide equivalent.
 Source: World Bank. 2023. [State and Trends of Carbon Pricing 2023](#).

5.4.2 Lessons for Economies in Asia and the Pacific

Developing an ETS should be an exhaustive process, with pilot projects and policy flexibility. This approach can be observed in the case of the PRC, which undertook ETS experimentation during the early 2010s, yielding pilot systems that were diverse and instructional. These early ETSs spanned the political and business hubs of Beijing and Shanghai, the sprawling industrial municipalities of Tianjin and Chongqing, the manufacturing locus of Guangdong province, the iron and steel center of Hubei province, and the special economic zone of Shenzhen. Based on the pilot schemes, the PRC's State Council released guidelines for accelerating the establishment of a national carbon market in 2017 and the nationwide ETS was introduced in 2021. With guidelines again updated in 2024, the PRC's national ETS is becoming a more impactful policy instrument (Gupta, Basu-Das, and Gupta 2024). Economies in Asia and the Pacific can learn from the PRC's progressive implementation and ever-evolving ETS, and some are taking similar approaches to increasing scope and ambition over time. For instance, Thailand's Greenhouse Gas Management Organisation has operated a voluntary ETS since 2013, focusing on the development and testing of monitoring, reporting, and verification (MRV) systems, setting caps and allocation procedures, and establishing trading infrastructure for 12 sectors with high GHG emissions. Meanwhile, Indonesia and Viet Nam both have three-phase plans that expand in coverage, compliance obligations, and potential impact through the 2020s. Prospective carbon markets elsewhere in South Asia and Southeast Asia will likewise go through progressive development stages if they are enacted.

The benefits and challenges of efficiency-based systems should be closely observed. The PRC, for example, has deployed efficiency-based targets in its ETS because such targets can be adapted to higher levels of economic production and aligned with broader PRC climate targets that lack an overall cap. Indonesia initiated an efficiency-based ETS in early 2023, focusing on power sector facilities with over 100 megawatts of production capacity. India's Perform, Achieve and Trade scheme may evolve into a compliance-based ETS and is also based on efficiency targets. It is likely that other developing economies in Asia and the Pacific will pursue this route, given their growth ambitions. However, while the impetus for efficiency-based targets is clear,

Box 5.3: Empirics of the European Union's Emissions Trading System

The emissions trading system (ETS) of the European Union (EU), a key instrument for decarbonizing the global economy, remains important to the learning experience across Asia and the Pacific.

The EU's ETS remains the world's largest cap-and-trade system, regulating greenhouse gas (GHG) emissions at over 11,000 installations—despite the ETS of the People's Republic of China being bigger in volume. The EU's ETS is also the oldest cap-and-trade system and the only multinational ETS comprising 30 individual economies (27 EU member states plus Iceland, Liechtenstein, and Norway). It is comprehensive in scope and covers about 45% of the EU's GHG emissions and is in its fourth trading phase (2021–2030). The EU's ETS was revised in 2023 in the context of the *European Green Deal* to align it with the target of reducing net emissions by 2030 to 62% below 2005 levels and net zero by 2050.

Since its launch in 2005, the EU's ETS has helped drive down emissions from electricity and heat generation and industrial production by 37.3% (European Commission 2023). It has also generated €184 billion (\$206 billion) in auction revenues for distribution to member states of which €43.6 billion was in 2023 alone (ICAP 2024). Member states have mostly used this revenue to fund investments in energy transformation and decarbonization as well as social measures in support of the green transition (European Commission 2023).

A growing body of literature points to significant positive impacts of the EU's ETS, based on economy-specific microdata studies (Colmer et al. 2022; Klemetsen, Rosendahl, and Jakobsen 2020; Petrick and Wagner 2014) and on sector-level data (Bayer and Aklın 2020). Further, a recent cross-study of installation-level data in France, the Netherlands, Norway, and the United Kingdom found that introduction of the EU's ETS was associated with a statistically significant reduction of carbon dioxide emissions in the order of –10% in the first two trading phases from 2005 to 2012. In addition, firm-level data on all ETS-regulated economies show the EU's ETS has had no significant impact on profits or employment levels and has led to an increase in revenues and fixed assets for regulated firms (Dechezleprêtre, Nachtigall, and Venmans 2023).

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Source: Gupta, S., S. Basu-Das, and S. Gupta. 2024. Carbon Pricing Through Emissions Trading In Asia-Pacific: A Comparative Analysis and the Way Forward. Background paper for the *Asia-Pacific Climate Report 2024: Catalyzing Finance and Policy Solutions*. Asian Development Bank.

such targets may limit the ability of an ETS to significantly curb absolute emissions. ETS designs should therefore seek to reduce these risks through broad coverage and steadily tightening benchmarks. Economies should incorporate clear timelines and strategies for ensuring the emissions impacts of their ETSs and for transitioning to mass-based systems.

Institutional and regulatory frameworks are pivotal to the smooth operation of an ETS. When developing their ETSs, economies of the Asia and Pacific region should dedicate significant resources to developing and refining the MRV systems that strengthen data management and accuracy. Legal mandates should be developed for facilities to report their GHG emissions data to ensure accurate emissions tracking and reporting, which will enhance system credibility and environmental integrity and lead to effective enforcement.

ETS institutional and regulatory frameworks are still being developed in many of the region's economies, as are the underpinning MRV systems. Policymakers charged with ETS portfolios need to provide clear mandates alongside roles and responsibilities to subsidiaries (e.g., state, provincial, and municipal governments) and other relevant bodies (e.g., registries, exchanges, third-party verifiers). To ensure smooth and effective ETS implementation, these stakeholders require predictable resources and staffing as well as substantial (and ongoing) capacity building and technical assistance from regional and international experts.

Reforming the energy sector to enable carbon price pass-through may be essential for an effective ETS. The energy sector in many economies in Asia and the Pacific is heavily regulated or in the early phases of liberalization. Regulated energy sectors hinder carbon price pass-through, negating the overall impact of an ETS. The PRC has for decades attempted varying levels of energy sector reform and, in 2020, it began a trial operation of eight pilot spot markets that utilize real-time electricity trading, formulate market-clearing prices, and guide electricity production and consumption through flexible market price signals. These spot markets have introduced price signals into electricity generation and distribution and, in doing so, they serve as an outlet for ETS cost pass-through to consumers. Their expansion, which will take time, along with further regulation and capacity building, could meaningfully improve the ability of the PRC's national ETS to reduce emissions. The PRC's experience shows that ETS architects and policymakers need to coordinate closely with power market reformers to promote a positive policy intersection, such as that created by the development of spot markets, enabling greater ETS effectiveness.

To add flexibility, countries can also allow for offset use in an ETS. However, such allowances should only be made to the extent that they do not undermine the effectiveness of the ETS. An ETS is inherently a flexible instrument because regulated entities can either purchase allowances from the market or reduce their own emissions, but allowing the use of offsets can provide additional flexibility. These offsets are generated through a crediting mechanism, which allows for the remuneration of achieved emissions reductions through the issuance of tradable certificates, while ensuring adherence to specific requirements. The crediting mechanism can be either domestic or international (section 5.5). Domestic offsets have been used in a range of ETS jurisdictions, e.g., subnationally in California and under the Regional Greenhouse Gas Initiative (encompassing Connecticut, Delaware, Maine, Maryland, Massachusetts, New Hampshire, New Jersey, New York, Pennsylvania, Rhode Island, and Vermont) and nationally in the PRC (Box 5.4). International offsets have historically played a greater role in Europe, the ROK, and New Zealand.

5.4.3 Emissions Trading System Linkages Across Economies

An international carbon pricing framework is needed to unify and achieve climate goals. Efforts to reduce emissions are fragmented due to divergent national climate ambitions and largely isolated ETSs. This highlights the need for an internationally coordinated carbon pricing system to achieve climate goals cost-effectively. The primary advantage of linking carbon markets is lowering emissions reduction costs (Anger 2008; Li and Duan 2021). Due to disparate marginal abatement costs, the cross-border exchange of carbon allowances can lower the costs of emissions compliance for all parties involved (Jaffe and Stavins 2008; Dellink et al. 2014). Economies exporting carbon permits benefit from sales revenue, while those importing permits benefit from reduced compliance costs (Qi and Weng 2016; Ranson and Stavins 2016). The basic concept of carbon market linkage is presented in Box 5.5.

Linked markets can enhance price discovery, liquidity provision, and risk management. Introducing more participants into carbon markets increases market liquidity and allows carbon prices to better reflect public information (Brunnermeier and Pedersen 2009; Santikarn, La Hoz Theuer, and Haug 2018; Brogaard, Hendershott, and Riordan 2019). This, in turn, contributes to the mitigation of price volatility (McKibbin, Shackleton, and Wilcoxon 1999; Jaffe and Stavins 2008; Flachsland, Marschinski, and Edenhofer 2009).

Box 5.4: A Domestic Voluntary Carbon Market—The China Certified Emission Reduction Scheme

Established in 2012, the China Certified Emission Reduction (CCER) scheme is central to the voluntary carbon market in the People's Republic of China (PRC) and plays a key role in the country's market-based carbon reduction strategy.

The CCER scheme features a domestic carbon credit offset trading standard used to monitor, report, and verify emission mitigation projects within the PRC. Carbon credits generated from a diverse range of projects—including renewable energy generation, methane capture, and forestry initiatives—have been traded in the CCER scheme, which provides avenues for businesses and individuals to voluntarily offset their emissions. Meanwhile, firms required to participate in the national compliance markets are allowed to purchase offset credits from the scheme at up to 5% of their quotation. Although the price level of CCER credits can sometimes fluctuate greatly, it is generally lower than that of the compliance market, so the scheme offers firms a flexible, often cost-effective way to achieve their emissions reduction targets.

Besides complementing the compliance market to achieve climate change mitigation, the CCER scheme helps channel substantial revenue from broad sources to targeted sectors such as renewable energy and forestry, which tend to be underinvested in a free market. For example, forestry projects under the CCER framework contribute to afforestation and reforestation, resulting in carbon sequestration and biodiversity conservation.

The CCER scheme also has the potential to become part of an international market, although it currently only operates within the PRC. It has largely adopted methodologies approved by the Clean Development Mechanism under the United Nations Framework Convention on Climate Change, and these methodologies are eligible for use with the Verified Carbon Standard program, the world's most widely used greenhouse gas crediting program.

The revamped CCER scheme, which was launched in early 2024, is the result of foresight by the Government of the PRC, which in 2017 suspended the issuance of new CCER credits in order to usher in a new era of effectiveness. The government recognized the need to streamline regulatory oversight, improve the quality of accredited projects, and reduce fraudulent claims and double-counting of emission reductions, through standardization and enhanced monitoring, reporting, and verification.

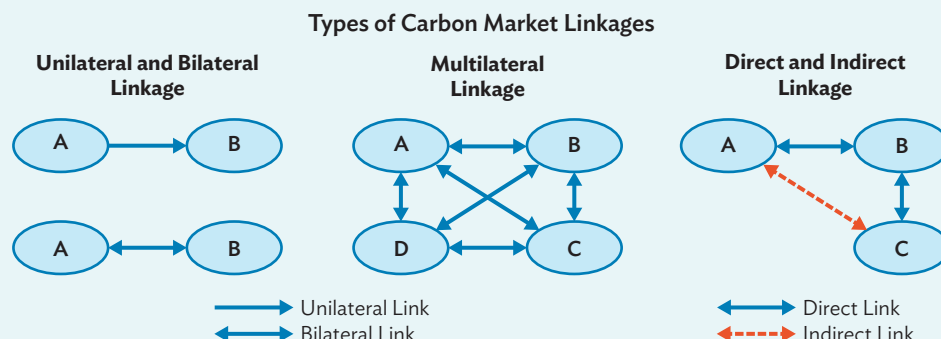
While keeping the strong features of the earlier scheme (e.g., its connection to the compliance market and use of internationally recognized methodologies), the new CCER scheme offers improved integrity, quality, and alignment with national climate policies. First, it expands the types of carbon offset projects allowed, enabling more clean energy projects to be registered and granted tradable CCER offset credits. Second, in line with the PRC's focus on integrating environmental and climate policies under a single authority, oversight of the new scheme has been shifted to the Ministry of Ecology and Environment, with this move expected to streamline regulatory protocols and enhance integration between the compliance and voluntary markets. Third, the new CCER scheme emphasizes transparency in reports for design, validation, verification, and data, which is expected to ensure consistency and stabilize prices.

These reforms to the CCER scheme are a significant milestone in the PRC's carbon reduction strategy, with the improvements attracting a wider range of participants and fostering greater confidence in both the domestic and international voluntary carbon markets.

Source: Gong, Y. 2024. The Voluntary Carbon Market, Globally and in the PRC: History, Current Situation and Prospects. Background paper for the *Asia-Pacific Climate Report 2024: Catalyzing Finance and Policy Solutions*. Asian Development Bank.

Box 5.5: Concept of Carbon Market Linkage

Carbon markets are considered linked if a participant in one market can use a carbon unit issued under another market to meet compliance obligations (Haïtes 2004; Haïtes 2016). Linkage between carbon markets can assume various forms, with differences in the degree, scope, and direction of trading flows, as shown in the figure below.



Source: Mehling, M. 2024. Theories and Practices of Cross-Border Carbon Market Linkages: Implications for Asia. Background paper for the Asia-Pacific Climate Report 2024: Catalyzing Finance and Policy Solutions. Asian Development Bank.

Direct links allow trade both within individual systems and between different systems (Ellis and Tirpak 2006). These links can be unilateral, bilateral, or multilateral. Bilateral or multilateral linkages offer the greatest economic benefits by increasing market size and diversity, enhancing integration efficiency, and leveling compliance costs through price convergence (Mehling and Haïtes 2009). A direct link can also be purely unilateral, which is appealing because it can be implemented without prior negotiations or mutual commitments.

An indirect linkage occurs when one carbon market links to a second system which, in turn, is linked to a third. Though not directly connected, the first and third markets influence each other through their mutual link to the second market, affecting unit supply, demand, and prices (Mehling and Haïtes 2009). Indirect links have commonly arisen when different markets accept the same offset credits, such as Certified Emission Reductions or Emission Reduction Units from the Kyoto Protocol mechanisms (Dellink et al. 2014; ADB 2016). Similarly, carbon credits authorized for international use under Article 6 of the Paris Agreement on Climate Change can be used to create indirect links.

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 Mehling, M. and E. Haïtes. 2009. *Mechanisms for Linking Emissions Trading Schemes*. Climate Policy. 9 (2). pp. 169–184. Background paper for the Asia-Pacific Climate Report 2024: Catalyzing Finance and Policy Solutions. Asian Development Bank.

Source: Mehling, M. 2024. Theories and Practices of Cross-Border Carbon Market Linkages: Implications for Asia. Background paper for the Asia-Pacific Climate Report 2024: Catalyzing Finance and Policy Solutions. Asian Development Bank.

ETS linkage can reduce “carbon leakage.” Due to the varying strictness of climate policies across different economies, regulated firms may consider relocating to jurisdictions with less-stringent regulations (a phenomenon known as carbon leakage), which undermines global efforts to reduce GHG emissions. Notably, the convergency of carbon prices in linked ETSs mitigates this risk (World Bank 2015).

Early unilateral carbon market linkages proved easy to establish and terminate. Carbon market linkages began in Europe and North America in the early 2000s. The Chicago Climate Exchange allowed Certified Emission Reductions in 2003 and allowances from the EU's ETS in 2005. However, these moves saw minimal uptake, leading to termination (Haites 2015). Norway linked its market unilaterally to the EU's ETS in 2005, fully integrating bilaterally in 2012. ETS markets in the EU, Japan, the ROK, New Zealand, Norway, and Switzerland linked project-based mechanisms under the Kyoto Protocol.¹⁴ The Tokyo Metropolitan Government linked its ETS with Saitama prefecture's in 2011. These experiences showed unilateral linkages are easy to start and end, while linkage to international credits decreased due to concerns over additionality, ensuring that emissions reductions would not have occurred with the reduction activity and the environmental integrity of Clean Development Mechanism units.

Successful long-term linkages rely on harmonization, transparency, support, and trust. Two prominent examples demonstrate that long-term ETS linkages rely on mutual recognition, regulatory harmonization, transparency, and coordinated administrative support to maintain market integrity and stakeholder trust. As the first example, California and Québec linked their ETSs in 2014. Ontario joined in 2017, but withdrew in 2018 after a change of government (Raymond 2020). Despite Ontario's withdrawal, California and Québec continued joint auctions and market activities, emphasizing transparency and coordinated administrative support for effective market management. As the second example, the EU and Switzerland signed an agreement to link their ETSs on 23 November 2017, effective 1 January 2020. The agreement includes mutual recognition of allowances, technical standards, and protection of sensitive information. It also established a joint committee for administration and dispute resolution and allows potential linkages with third parties. Transactions between the EU and Swiss registries became operational in 2021, ensuring system integrity and transparency.

Differences in institutional settings, climate goals, and policy objectives add significant challenges.

While procedural disparities—such as sectoral and GHG coverage and allowance allocation methods—can be harmonized, differences in legal frameworks and climate policy legislation can impede ETS consensus among economies (Fankhauser, Gennaioli, and Collins 2015). Varying ambitions on climate goals also further complicate linkages. Economic circumstances may lead governments to scale back emissions reduction initiatives and relax enforcement standards. Effective cross-border carbon markets require harmonization among ETSs, which can conflict with domestic policy objectives (Flachsland, Marschinski, and Edenhofer 2009). Maintaining control over individual carbon market policies also poses a challenge, as bilateral and multilateral linkages necessitate high consistency in cap settings, allowance allocation, and MRV processes.

A focus on domestic markets can provide a platform for future cross-border linkages. Given the complexity of establishing an extensive cross-border carbon market linkages, each compliance market should prioritize further development of domestic markets while aiming to leverage the benefits from carbon market connectedness. To facilitate the linking process, the following six policy recommendations are suggested: (i) establish a communication forum or leverage international organizations to share policies and coordinate efforts, ensuring transparency and consistent information flow; (ii) implement a legally-backed MRV framework with enforcement measures to ensure accurate emissions tracking and reporting, enhancing system credibility and environmental integrity; (iii) create an authorized cooperative organization and robust trading infrastructure for carbon-related products, including post-trade registries; (iv) ensure accurate measurement of international carbon allowances to mitigate double-issuing and double-using of carbon credits; (v) implement cooperative policies to maintain environmental integrity, promote common objectives such as nationally determined contributions, and incentivize future mitigation actions; and (vi) strengthen financial regulations and policies to mitigate risks—such as fraud, misleading information, manipulation, and transfer pricing—in carbon finance.

¹⁴ Since 2020, the EU no longer accepts the [use of international credits](#), including Certified Emission Reductions, for compliance under its ETS.

5.5 The Role of International Carbon Markets

5.5.1 Recent Development of International Carbon Markets

There is an emerging landscape of international carbon markets for cost-effective emissions reduction.

International carbon markets enable the trading of carbon credits internationally, both through the compliance market under Article 6 of the Paris Agreement and through voluntary carbon markets (VCMs). These markets are based on baseline-and-crediting mechanisms, whereby carbon credits are generated through investments that reduce emissions relative to a counterfactual “baseline” estimate of future emissions. Independent carbon crediting programs provide systems for the generation of carbon credits, as does Article 6.4, also known as the Paris Agreement Crediting Mechanism (PACM). These markets are important as they mobilize carbon finance, which can enhance the attractiveness of private investment in low-carbon technologies and solutions. Governments and private companies can therefore utilize international carbon markets to reduce emissions cost-effectively and access funding for sustainable development.

Article 6 lays the foundation for international compliance markets under the Paris Agreement. Article 6 enables both developing and developed economies to buy and sell Internationally Transferred Mitigation Outcomes (ITMOs) to achieve their mitigation commitments under nationally determined contributions (NDCs). Within Article 6, international cooperation on climate mitigation can be pursued through two market-based approaches established under Article 6.2 and Article 6.4, respectively. Article 6.2 is a decentralized approach that offers greater flexibility in defining the scope and methods of cooperation while using common accounting and reporting rules. For example, under Article 6.2, market scope and methods can be defined bilaterally between cooperating Parties while complying with the guidance for accounting and reporting from the United Nations Framework Convention on Climate Change (UNFCCC). Article 6.4, now termed the PACM, uses a centralized approach and is governed by a supervisory body of the UNFCCC. Since the adoption of the Paris Agreement, significant progress has been made in clarifying how these market mechanisms are intended to function. Under Article 6 rules, decisions have been taken that the trading of ITMOs under the Paris Agreement will require authorization from the host government and the application of a corresponding adjustment to avoid double-counting with host economy NDCs. However, additional decisions are needed to reduce uncertainty for Article 6.2 and to operationalize the PACM.

International carbon markets under Article 6 can generate significant global financing. While estimates of such financing fall within a wide range, studies suggest the potential economic benefits of Article 6 cooperation are significant. The World Bank estimates that resource flows generated through international carbon trading, under full and cost-effective implementation of NDC targets, need to be \$185 billion in 2030 (World Bank 2016). The International Emissions Trading Association notes Article 6 market projections of \$250 billion a year by 2030 and \$1 trillion a year by 2050 (Edmonds et al. 2019).

While international markets are still developing, economies in Asia and the Pacific are already active. As of 5 July 2024, there were 83 bilateral carbon trading agreements between 10 different buyers and 46 host economies globally, with 140 pilot projects recorded (of which 119 belong to Japan’s Joint Crediting Mechanism) (UNEP 2024). In Asia and the Pacific, Thailand and Vanuatu have provided national authorization for the export of credits under Article 6 and the first-ever issuance of 1,916 ITMOs occurred for the E-Bus Program in Bangkok in 2023: this transaction between Thailand and Switzerland is the first under Article 6.2 (UNEP 2024).

The most immediate opportunity for issuance of carbon credits through the PACM relates to the transition of Clean Development Mechanism (CDM) activities. Parties to the Paris Agreement agreed that projects meeting eligibility requirements should be able to transition from the CDM to the PACM and thus continue to generate carbon credits. According to an assessment by the United Nations Environment

Programme, as of 5 July 2024, transition requests were correctly submitted for 1,356 project activities and 118 programs of activities (including 949 component project activities). Total potential emissions reductions for 2021–2025 generated from transition CDM activities could amount to more than 900 million metric tons of CO₂ as per the established CDM methodologies (UNEP 2024).

Climate-conscious enterprises can benefit greatly from voluntary carbon markets. International VCMs can be utilized by organizations, typically in the private sector, to support achieving their own voluntary mitigation commitments, such as pledges to achieve carbon neutrality or net zero emissions. International VCMs encompass a collection of standard bodies, voluntary buyers, and carbon credit generators that sell verified carbon credits to be used for voluntary purposes. Independent carbon crediting programs or standards, such as the Gold Standard and VERRA's Verified Carbon Standard, set the requirements for delivering and measuring emissions reductions and removals resulting from a mitigation activity.

VCMs must rebound from falls in trading values and volumes in 2022 and 2023. The total trading value of all VCMs (international) peaked in 2021, reaching about \$2.1 billion, then declined to \$1.9 billion in 2022 and to \$0.72 billion in 2023. Transaction volumes of VCM-traded carbon credits in metric tons of carbon dioxide equivalent (tCO₂e) also peaked at 516 million in 2021, then fell to 254 million in 2022 and to 111 million in 2023. The reported average nominal price in 2023 was \$6.53 per tCO₂e, an 11% decline on 2022, but over 60% above the reported average nominal price in 2021 (Procton 2024). There is a mix of concerns regarding VCMs, particularly their high transactions costs and demand uncertainty. There are also issues regarding overall market integrity related to the varying quality of credits, due to inflated baselines and challenges of monitoring, as well as negative impacts on local communities. However, despite trading declines and obstacles hindering growth, the overall sentiment for VCMs is optimistic.

Efforts are being made to address integrity issues in VCMs. Two notable initiatives for voluntary markets are the Integrity Council for the Voluntary Carbon Market (ICVCM) and the Voluntary Carbon Markets Integrity Initiative (VCMI). These two bodies are establishing guidelines and standards for high-integrity VCMs on the supply side and demand side, respectively. The publication of the ICVCM's Core Carbon Principles and the launch of the VCMI's Claims Code are contributing to buyer confidence in market quality and integrity. In addition to embracing ICVCM and VCMI guidelines, governments are taking their own measures to build confidence in VCMs. For example, the Government of the United States has announced new principles to advance the responsible development of VCMs, thereby encouraging these markets to embrace identified approaches for better supply, demand, and market-level integrity. Papua New Guinea, meanwhile, is regulating nature-based solutions projects in particular, with a benefit-sharing mechanism as well as stringent, free, prior, and informed content provisions.

Industry-specific approaches can complement broader international carbon schemes. Alongside Article 6 and the VCMs, there are industry-specific approaches that influence international carbon markets. One example is the Carbon Offsetting and Reduction Scheme for International Aviation. International aviation is not included in national emissions inventories nor in NDCs, and has instead been addressed by the International Civil Aviation Organization (ICAO). In 2016, the ICAO Assembly adopted a global market-based emissions trading scheme for international aviation. A technical committee under the ICAO has determined eligibility criteria for carbon credits, allowing independent carbon crediting programs, the PACM, and domestic carbon crediting programs to generate carbon credits. The credits will need authorization under Article 6.2 of the Paris Agreement (Anjaparidze 2019).

5.5.2 Opportunities and Challenges for Asia and the Pacific

Developing economies can use international carbon markets to meet domestic climate goals. With the finalization of the Paris Agreement Rulebook, many countries are operationalizing international carbon markets under Article 6 as well as engaging with VCMs. Although developing economies in Asia and the Pacific are mostly exploring avenues to export carbon credits, there is also potential for them to meet their climate ambitions by buying credits through international carbon markets. This could occur when an economy has projected that it may not reach its NDC targets (especially when these targets have been recently lifted) or when purchasing credits proves to be cheaper than domestic mitigation efforts (which may have high marginal abatement costs). Assuming that using public funds to purchase carbon credits is politically challenging, this task can be delegated to the private sector. For instance, economies with a high domestic carbon price may decide to tighten the cap of their ETS but allow for private entities under the cap to purchase international carbon credits (ITMOs), since this may constitute a more cost-effective alternative to high ETS prices. This has been the case in Indonesia.

The Asia and Pacific region has extensive experience in carbon market mechanisms. Approximately 80% of all CDM projects are based in Asia and the Pacific (Amarjargal et al. 2020). While project developers and validation and verification bodies can draw on experiences from the CDM, host economies have less to build on since Article 6 requires greater host involvement and higher institutional and MRV requirements. Japan's Joint Crediting Mechanism (JCM) is a forerunner to Article 6.2 and around 90% of all JCM projects are hosted in the region (IGES 2024). Developing economies in Asia and the Pacific can draw on experiences from the JCM as a bilateral approach.

The region has work to do in scaling up involvement in international carbon markets. Some of the existing challenges include: (i) lack of national strategies and frameworks for engagement with international carbon markets; (ii) ineffective governance systems including lack of interministerial coordination as well as lack of mandates and legal instruments; (iii) lack of institutional capacity and/or skills of personnel to negotiate bilateral agreements; and (iv) lack of the carbon market expertise required to manage domestic processes and procedures for engagement in international markets.

National carbon market strategies are a vital catalyst for international engagement. Alongside enhancing institutional capacity and technical readiness to participate in international carbon markets, an important first step for many economies is to develop holistic national carbon market strategies (ADB 2023b). To engage strategically in carbon markets, policymakers need to weigh the benefits of selling mitigation outcomes to attract carbon finance against the need to meet their NDC targets. Evaluating this trade-off requires a long-term strategy that aims to achieve NDC targets and informs engagement with Article 6, including guiding principles and criteria for mitigation outcomes to be eligible for international transfer. Such a strategy must embrace the interrelationship between various carbon markets and cover management of both the compliance and voluntary markets. For instance, buyers in a VCM may wish to have carbon credits (ITMOs) that have authorization and cannot be used toward a host economy's NDC targets. Another type of integration can occur when domestic carbon crediting mechanisms are used for Article 6.2 transactions, which may require new regulations and legal frameworks.

The NDC accounting aspect of Article 6 is critical to effective market integration. For economies to engage with international carbon markets, they need to have a framework outlining the sectors and types of mitigation activities from which they will allow ITMO exports. This, in turns, requires an understanding of how the NDC system will be implemented. The framework also needs to describe administrative procedures to manage ITMO authorization, tracking of ITMOs, and their accounting (corresponding adjustments). Governments may consider what actions need to be taken to promote voluntary market activities within their jurisdiction and may consider regulating such markets.

5.5.3 Support for Nature-Based Solutions Through International Carbon Markets

Nature-based solutions (NBSs) have the potential to ramp up global climate finance. As global efforts intensify to achieve net zero emissions by 2050, integrating NBSs into international carbon markets offers a pathway to mobilize financial resources at scale. These solutions leverage the power of nature and healthy ecosystems to protect people, optimize infrastructure, and safeguard a stable and biodiverse future (Cohen-Shacham et al. 2016). They include activities such as climate-smart agriculture, biodiverse habitat restoration, water resource management, clean cookstove projects, and green infrastructure initiatives. A significant aspect of NBSs is their long-term benefits. For instance, sustainable land-use change can mean soils continue to store CO₂ even after global emissions peak, reducing temperatures in the long term and contributing to planetary cooling. In fact, NBSs have the potential to provide up to 37% of the mitigation necessary by 2030 to reach the targets of the Paris Agreement (Griscom et al. 2017).

Despite these benefits, NBSs face a significant financing gap. Required investment in NBSs stood at \$200 billion annually in 2022 and needs to triple by 2030 and quadruple by 2050 to reach climate, biodiversity, and land degradation targets (UNEP 2023). The primary challenge lies in the undervaluation of nature's benefits within traditional economic frameworks and conventional financing systems, leading to insufficient private investment. Higher risks, lower returns, and longer payback periods discourage investment in NBSs, despite their potential to enhance climate and ecosystem resilience (UNEP 2023). Public finance therefore has been the primary source of finance flows to NBSs, accounting for 82% in 2023, with a focus on biodiversity, landscape protection, sustainable agriculture, forestry, and fishing. Private finance accounts for only 18% of total finance flowing to NBSs, with more than half channeled through biodiversity credits, offsets, and sustainable supply chains (UNEP 2023).

Carbon markets present an opportunity to address this gap. They can do so by assigning a financial value to carbon sequestration and emission reductions through nature. It is estimated that NBSs financed through carbon markets can deliver 10%–12% of the mitigation needed by 2030 to be on track to reach global net zero by 2050 (Landholm et al. 2022). These estimations account for price, implementation feasibility, and spatial location constraints.

Agriculture, forestry, and wetlands management can be driving forces in carbon markets. The core potential of NBSs in carbon markets is projected to be in the agriculture sector, forestry (avoided deforestation,¹⁵ afforestation and/or reforestation, and improved forest management), and the restoration and conservation of wetlands. Using 2030 as the benchmark, agriculture at 43% has the highest carbon market potential under NBS measures, followed by avoided deforestation at 32%, afforestation and/or reforestation at 11%, improved forest management at 7%, and the restoration and conservation of wetlands at 7% (Landholm et al. 2022). The top five economies with the highest potential in agriculture carbon markets are the United States, the PRC, India, Brazil, and Indonesia. The potential for avoided deforestation is largest in Latin America and Asia, particularly in Brazil, Indonesia, Colombia, Peru, and Malaysia. Considering land mass as the measure, the largest mitigation potential across all NBS carbon market activities lies in Indonesia, Malaysia, Costa Rica, Colombia, and Belize (Landholm et al. 2022).

¹⁵ Negotiations at the 60th session of the Subsidiary Body for Scientific and Technological Advice (SBSTA 60) in Bonn, 3–13 June 2024, included draft decisions to refrain from carbon crediting of avoided deforestation emissions until 2028, when the issue will be discussed again. This implies that avoided deforestation as a mitigation activity is only eligible on the VCMs.

Prices for NBS carbon credits vary depending on the solution type, project, and market used. However, they typically range from \$5 to \$15 per tCO₂e, reflecting the market's recognition of their interrelated environmental and social benefits (Procton 2024). As global climate ambitions grow, demand for high-quality NBS credits is expected to rise, creating new opportunities for investment and innovation.

The future of NBSs in international carbon markets appears complex yet promising. Despite its high potential, the NBS market must overcome several challenges, including the need for robust MRV systems to ensure the integrity and transparency of the carbon credits and for protocols that can underpin the permanence of NBS efforts. Policy frameworks must evolve to support the scaling up of NBS projects. This includes addressing barriers such as land tenure issues, regulatory complexities, and access to high-integrity carbon finance for local communities. Nevertheless, integrating NBSs into international carbon markets offers a promising means by which to mobilize resources for climate mitigation and sustainable development. By valuing ecosystem services and carbon sequestration through market mechanisms, carbon markets can incentivize investment in NBS projects that deliver environmental, social, and economic benefits.

5.6 The Effects of Carbon Pricing on Developing Economies

Despite strong momentum toward carbon pricing, there are some concerns for developing economies.

In particular, there has been speculation that efforts to implement carbon pricing may lead to a loss of global competitiveness. An additional concern is the risk of carbon leakage in response to carbon pricing, with firms shifting production out of certain jurisdictions to avoid paying a carbon price, thereby reducing the ability of governments to regulate GHG emissions.

The impact of carbon pricing on competitiveness is ambiguous. While the imposition of carbon pricing will increase costs for local firms that are subjected to the carbon price, it can also be an incentive to modernize production techniques to drive improvements in productivity and competitiveness. Consistent with the Porter Hypothesis, a carbon price can also encourage innovation in new green technologies, leading to improvements in competitiveness by avoiding the burden of carbon pricing and increasing both demand (through consumer preference for climate-friendly brands) and profits. Empirical evidence suggests that the competitiveness effects of carbon pricing are usually quite small (see, for example, Arlinghaus 2015). However, studies of existing carbon pricing schemes have been undertaken in the context of low carbon prices and may not reflect effects of carbon pricing at higher levels.

The overall impact also depends on the global coverage of carbon pricing schemes. With carbon pricing not occurring globally, there is the risk that firms operating in economies with carbon pricing may lose market share and see reduced profits, with production shifting to jurisdictions with no carbon pricing (or where carbon is priced lower). The extent and effect of this carbon leakage will be diminished with increased coordination and coverage of carbon pricing schemes across the world, because it will become more difficult for firms to shift production beyond carbon pricing boundaries.

More is revealed about competitiveness impacts when advanced modeling is applied. Adopting a computable general equilibrium model allows for an examination of the competitiveness impacts of carbon pricing under different pricing and coverage scenarios. The modeling exercise starts with a scenario that examines the impact of the implementation of the Carbon Border Adjustment Mechanism (CBAM) by the EU (Table 5.2). The CBAM is intended to provide surety for EU firms by charging an equivalent carbon price on imported intermediate inputs as on domestic inputs. Given the potential impact of this initiative on the

competitiveness of economies in Asia and the Pacific, the initial scenario assumes the imposition of the CBAM and an increase in the EU's ETS carbon price to €100 per metric ton of CO₂, from a baseline of €18.¹⁶ With other developed economies likely to follow the EU in implementing Border Carbon Adjustment (BCA) mechanisms, a second scenario involves extending the CBAM to all Organisation for Economic Co-operation and Development (OECD) economies (including those in Asia and the Pacific). The remaining scenarios implement carbon pricing, including BCA, in specific subregions of Asia and the Pacific to examine the estimated competitiveness effects of carbon pricing on these subregions, allowing for substitution effects toward subregions without carbon pricing. To account for the additional challenges for least-developed economies, additional scenarios are introduced that involve these economies charging a lower carbon price of €50. The final scenario introduces carbon pricing across all of Asia and the Pacific, with the main substitution effects then being toward other regions of the world.

Table 5.2: Overview of Carbon Pricing Scenarios

<i>Scenario 1</i>	- European economies impose tighter ETS carbon allocations, with a resulting €100/MT price. - CBAM taxes are imposed for ETS sectors.
<i>Scenario 2</i>	- All OECD economies impose tighter ETS carbon allocations, with a resulting €100/MT price. - CBAM taxes are imposed for ETS sectors.
<i>Scenario 3</i>	- OECD economies impose tighter ETS carbon allocations, with a resulting €100/MT price. - Non-LDE SE Asian countries impose tighter ETS carbon allocations, with a resulting €100/MT price. - CBAM taxes are imposed for ETS sectors.
<i>Scenario 4</i>	- OECD economies impose tighter ETS carbon allocations, with a resulting €100/MT price. - Non-LDE East Asian economies impose tighter ETS carbon allocations, with a resulting €100/MT price. - CBAM taxes are imposed for ETS sectors.
<i>Scenario 5</i>	- OECD economies impose tighter ETS carbon allocations, with a resulting €100/MT price. - Non-LDE South Asian countries impose tighter ETS carbon allocations, with a resulting €100/MT price. - CBAM taxes are imposed for ETS sectors.
<i>Scenario 6</i>	- OECD economies impose tighter ETS carbon allocations, with a resulting €100/MT price. - Non-LDE Other Asian countries impose tighter ETS carbon allocations, with a resulting €100/MT price. - CBAM taxes are imposed for ETS sectors.
<i>Scenario 7</i>	- OECD economies impose tighter ETS carbon allocations, with a resulting €100/MT price. - All non-LDE Asian countries impose tighter ETS carbon allocations, with a resulting €100/MT price. - CBAM taxes are imposed for ETS sectors.
<i>Scenario 8</i>	- OECD economies impose tighter ETS carbon allocations, with a resulting €100/MT price. - All non-LDE Asian countries impose tighter ETS carbon allocations, with a resulting €100/MT price. - All LDE Asian countries impose tighter ETS carbon allocations, with a resulting €50/MT price. - CBAM taxes are imposed for ETS sectors.

CBAM = Carbon Border Adjustment Mechanism, ETS = emissions trading system, LDE = least-developed economy, MT = metric ton, OECD = Organisation for Economic Co-operation and Development.

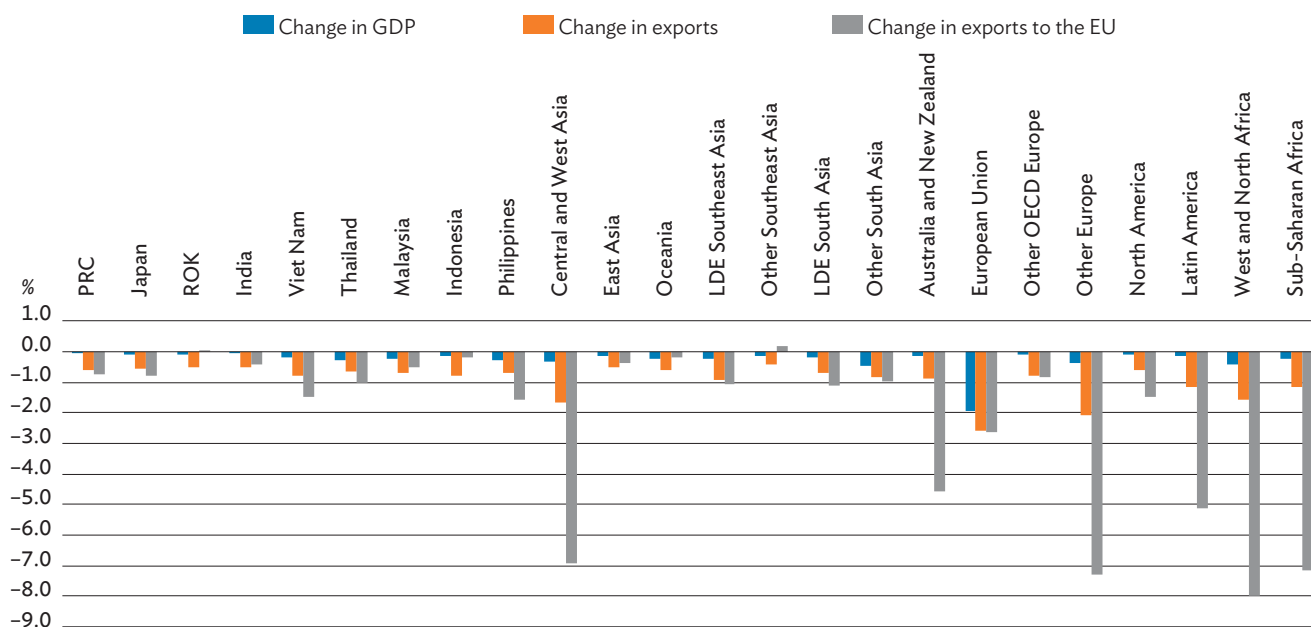
Notes: During the phase-in period, the CBAM regime will not apply to all ETS sectors. However, the CBAM system is expected to be expanded to all ETS sectors after the phase-in period. Central and West Asia comprises Armenia, Azerbaijan, Georgia, Kazakhstan, the Kyrgyz Republic, Pakistan, Tajikistan, Turkmenistan, and Uzbekistan. LDE South Asia comprises Bangladesh and Nepal. LDE Southeast Asia comprises Cambodia, the Lao People's Democratic Republic, Myanmar, and Timor-Leste. Other South Asia comprises Bhutan, India, Maldives, and Sri Lanka. Other Southeast Asia comprises Brunei Darussalam, Indonesia, Malaysia, the Philippines, Singapore, Thailand, and Viet Nam.

Source: Asian Development Bank.

¹⁶ The baseline of €18 per metric ton of CO₂ reflects the approximate price of CO₂ in the reference year of 2017. The revenue collected from the CBAM is assumed to go into the EU's budget in the model.

Imposing the CBAM and increasing the EU carbon price lowers estimated global output levels and exports. This outcome is consistent with strong income effects, with the higher costs of production in the EU leading to lower global output. The EU is most strongly affected, with gross domestic product (GDP) estimated to fall by almost 2%. Effects in other regions are generally muted, with those regions closer to the EU (e.g., Other Europe, West and North Africa, and Central and West Asia) tending to see somewhat larger declines compared to other regions (Figure 5.4). Even in these cases, however, reductions in GDP are estimated at less than 0.5%. Impacts on exports tend to be greater, with reductions of more than 1% in the EU, Other Europe, North and West Africa, Sub-Saharan Africa, Latin America, and Central and West Asia.

Figure 5.4: Modeled Impacts of the Carbon Border Adjustment Mechanism and Changes to the European Union's Emissions Trading System



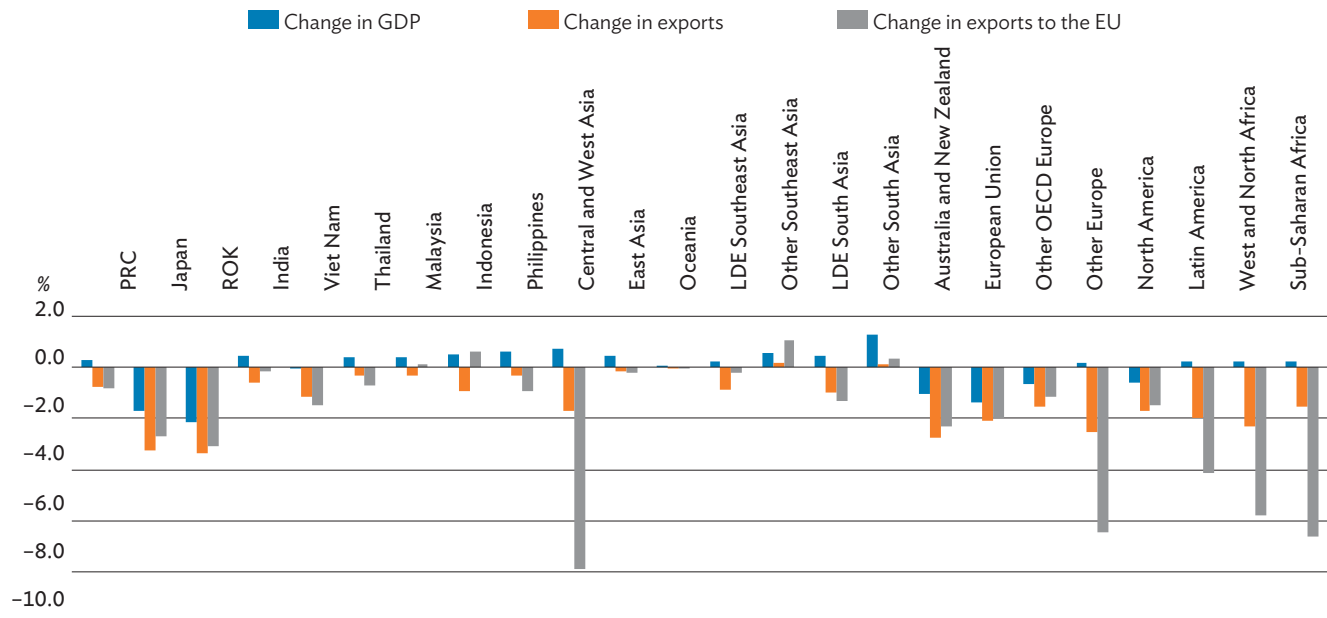
PRC = People's Republic of China, EU = European Union, GDP = gross domestic product, LDE = least-developed economy, OECD = Organisation for Economic Co-operation and Development, ROK = Republic of Korea.

Notes: Central and West Asia comprises Armenia, Azerbaijan, Georgia, Kazakhstan, the Kyrgyz Republic, Pakistan, Tajikistan, Turkmenistan, and Uzbekistan. LDE South Asia comprises Bangladesh and Nepal. LDE Southeast Asia comprises Cambodia, the Lao People's Democratic Republic, Myanmar, and Timor-Leste. Other South Asia comprises Bhutan, India, Maldives, and Sri Lanka. Other Southeast Asia comprises Brunei Darussalam, Indonesia, Malaysia, the Philippines, Singapore, Thailand, and Viet Nam.

Source: ADB calculations using data from the [Global Trade Analysis Project 11](#) and the [International Energy Agency](#) (both accessed 15 November 2023).

Extending the ETS and CBAM to the rest of the OECD has varied effects. Many economies in Asia and the Pacific have estimated increases in output (Figure 5.5). While negative effects on output and exports within the EU diminish, effects in other OECD regions (Japan, the ROK, Other OECD Europe, and North America) become larger, with reductions in output and exports being relatively large in Japan and the ROK. In many economies and subregions of Asia and the Pacific, however, output levels (and in some cases exports) are estimated to rise. This is especially true for Other South Asia, Central and West Asia, Other Southeast Asia, the Philippines, Indonesia, and Thailand. These changes likely reflect carbon leakage, with downstream production shifting out of the OECD and into these economies.

Figure 5.5: Modeled Impacts of the Carbon Border Adjustment Mechanism and Emissions Trading Across OECD Economies



PRC = People's Republic of China, EU = European Union, GDP = gross domestic product, LDE = least-developed economy, OECD = Organisation for Economic Co-operation and Development, ROK = Republic of Korea.

Notes: Central and West Asia comprises Armenia, Azerbaijan, Georgia, Kazakhstan, the Kyrgyz Republic, Pakistan, Tajikistan, Turkmenistan, and Uzbekistan. LDE South Asia comprises Bangladesh and Nepal. LDE Southeast Asia comprises Cambodia, the Lao People's Democratic Republic, Myanmar, and Timor-Leste. Other South Asia comprises Bhutan, India, Maldives, and Sri Lanka. Other Southeast Asia comprises Brunei Darussalam, Indonesia, Malaysia, the Philippines, Singapore, Thailand, and Viet Nam.

Source: ADB calculations using data from the [Global Trade Analysis Project 11](#) and the [International Energy Agency](#) (both accessed 15 November 2023).

Imposing carbon pricing selectively in subregions negatively impacts estimated output in those subregions.

When a particular subregion imposes carbon pricing, it suffers reductions in output. In the case of East Asia, for example, output is estimated to fall by more than 2.2% in the PRC and 1.2% in the rest of East Asia (Table 5.3).¹⁷ There is also strong evidence of carbon leakage in response to carbon pricing in subregions. Considering the case of East Asia, the imposition of carbon pricing is estimated to lead to substantial gains in GDP in other subregions and certain economies, with estimated increases of 4.6% in Other South Asia, 2.9% in Central and West Asia, 2.4% in the Philippines, 2.2% in Thailand, and 2.0% in Other Southeast Asia. Similar patterns are observed when looking at carbon pricing in other subregions.

Extending carbon pricing to all of Asia and the Pacific encourages carbon leakage and creates economic inequities.

A coordinated policy of carbon pricing across the Asia and Pacific region (excluding least-developed economies) is estimated to have two main effects. First, it encourages carbon leakage to other regions, with output in West and North Africa estimated to increase by 2.3%, and with relatively large increases in Latin America (1.4%), Other Europe (1.8%), Sub-Saharan Africa (1.4%), and North America (0.8%). Second, it creates inequities within Asia and the Pacific. Reductions in output are estimated for East Asia and Oceania (the Pacific) and in the PRC, India, Malaysia, Thailand, and Viet Nam. Meanwhile, Indonesia, the Philippines, Central and West Asia, Other and Low-Income Southeast Asia, and Other and Low-Income South Asia are estimated to see output increase. This will partially reflect differences in CO₂ intensities, with the carbon price shifting production to those with relatively low intensities. Similar patterns are found to hold when introducing a (lower-priced) ETS and the CBAM in least-developed economies of Asia. In this scenario, these economies are still estimated to see an increase in output from region-wide carbon pricing, though the extent of this

¹⁷ The approach assumes that both the EU and the OECD also impose an ETS and the CBAM, meaning that the reported effects also include the effects of carbon pricing in these regions.

increase is diminished by its own carbon price. Such results lead to an important conclusion: even uniform carbon pricing in the Asia and Pacific region can have distributional consequences, depending on the carbon intensity and structure of production of the different economies and subregions.

Table 5.3: Estimated Output Changes in Response to Carbon Pricing Throughout Asia and the Pacific (%)

	All OECD and Southeast Asia	All OECD and East Asia	All OECD and South Asia	All OECD and Other Asia and the Pacific	All OECD and All Non-LDE Asia	All OECD and All Asia
PRC	0.34	-2.15	0.47	0.32	-1.90	-1.89
Japan	-1.66	-0.33	-1.38	-1.66	0.17	0.18
Republic of Korea	-2.08	-1.06	-1.83	-2.08	-0.62	-0.60
India	0.51	1.33	-2.93	0.49	-1.92	-1.94
Viet Nam	-2.06	0.47	0.13	-0.01	-1.38	-1.32
Thailand	-4.87	2.20	0.81	0.44	-2.65	-2.45
Malaysia	-2.56	1.76	0.69	0.42	-0.85	-0.73
Indonesia	-0.52	1.90	0.83	0.56	1.21	1.30
Philippines	-1.56	2.37	1.11	0.73	0.72	0.80
Central and West Asia	0.80	2.87	1.23	-1.85	0.93	0.89
East Asia	0.51	-1.15	0.77	0.53	-0.72	-0.70
Oceania	-0.09	0.74	0.16	-2.56	-1.84	-1.80
LDE Southeast Asia	1.42	1.25	0.49	0.29	2.73	1.53
Other Southeast Asia	-1.46	1.96	0.86	0.58	0.20	0.38
LDE South Asia	0.47	1.68	0.61	0.49	1.97	1.00
Other South Asia	1.30	4.61	-3.24	1.46	0.29	0.22
Australia and New Zealand	-1.07	-0.66	-0.97	-1.03	-0.59	-0.58

PRC = People's Republic of China, LDE = least-developed economy, OECD = Organisation for Economic Co-operation and Development.

Notes: Central and West Asia comprises Armenia, Azerbaijan, Georgia, Kazakhstan, the Kyrgyz Republic, Pakistan, Tajikistan, Turkmenistan, and Uzbekistan. LDE South Asia comprises Bangladesh, Nepal. LDE Southeast Asia comprises Cambodia, the Lao People's Democratic Republic, Myanmar, Timor-Leste. Other Southeast Asia comprises Brunei Darussalam, Indonesia, Malaysia, the Philippines, Singapore, Thailand, and Viet Nam. Other South Asia comprises Bhutan, India, Maldives, and Sri Lanka.

Source: ADB calculations using data from the [Global Trade Analysis Project 11](#) and the [International Energy Agency](#) (both accessed 15 November 2023).

Estimated impacts of carbon pricing on exports are largely consistent with those for output.

The estimated impacts on those economies and subregions implementing carbon pricing are generally in line with those for output, with exports declining substantially (Table 5.4). However, for the other economies and subregions not subject to carbon pricing, there are often differences between the output and export effects. In the case of carbon pricing by East Asia, for example, the estimates suggest increases in output but reductions in exports for Central and West Asia, least-developed economies in Southeast Asia and South Asia, and in India, Indonesia, and Viet Nam. Such results highlight that, while carbon pricing in one subregion can lead to increased output in other subregions, this does not necessarily equate to an increase in exports and global competitiveness. This further suggests that, in these cases, the increased output is linked to serving a higher share of domestic demand.

Table 5.4: Estimated Changes in Exports in Response to Carbon Pricing Throughout Asia and the Pacific (%)

	All OECD and Southeast Asia	All OECD and East Asia	All OECD and South Asia	All OECD and Other Asia and the Pacific	All OECD and All Non-LDE Asia	All OECD and All Asia
PRC	-0.89	-2.09	-0.52	-0.77	-1.90	-1.89
Japan	-3.39	-2.25	-2.84	-3.18	-1.95	-1.95
Republic of Korea	-3.41	-2.66	-2.92	-3.30	-2.23	-2.24
India	-0.74	-0.21	-3.92	-0.55	-3.67	-4.07
Viet Nam	-2.32	-1.04	-1.01	-1.20	-2.16	-1.93
Thailand	-4.16	1.35	0.01	-0.40	-2.18	-2.00
Malaysia	-3.02	0.70	-0.13	-0.41	-1.81	-1.50
Indonesia	-2.26	-0.37	-1.27	-1.17	-2.33	-1.76
Philippines	-2.47	0.41	0.07	-0.30	-1.40	-1.14
Central and West Asia	-1.85	-1.37	-1.66	-2.42	-2.15	-3.01
East Asia	-0.21	-2.79	0.20	-0.08	-2.38	-2.39
Oceania	-0.25	1.10	0.16	-4.09	-3.02	-3.06
LDE Southeast Asia	0.46	-0.13	-0.75	-0.84	1.36	0.26
Other Southeast Asia	-2.78	1.53	0.32	-0.03	-1.44	-0.95
LDE South Asia	-1.13	-1.21	-1.85	-1.02	-2.22	-0.20
Other South Asia	-0.06	2.24	-2.44	0.23	-0.44	-0.70
Australia and New Zealand	-3.08	-3.41	-2.91	-2.80	-3.91	-3.90

PRC = People's Republic of China, LDE = least-developed economy, OECD = Organisation for Economic Co-operation and Development. Notes: Central and West Asia comprises Armenia, Azerbaijan, Georgia, Kazakhstan, the Kyrgyz Republic, Pakistan, Tajikistan, Turkmenistan, and Uzbekistan. LDE South Asia comprises Bangladesh, Nepal. LDE Southeast Asia comprises Cambodia, the Lao People's Democratic Republic, Myanmar, Timor-Leste. Other Southeast Asia comprises Brunei Darussalam, Indonesia, Malaysia, the Philippines, Singapore, Thailand, and Viet Nam. Other South Asia comprises Bhutan, India, Maldives, and Sri Lanka.

Source: ADB calculations using data from the [Global Trade Analysis Project 11](#) and the [International Energy Agency](#) (both accessed 15 November 2023).

Potential effects of carbon pricing call for greater coordination and new compensation mechanisms.

The results of the modeling exercise show that imposing carbon prices unilaterally can have negative consequences for output and exports. Many of these effects disappear with increased coordination and coverage of carbon pricing, though distributional effects still result in economic inequities for economies and subregions. Such results highlight the need for increased efforts at global cooperation in encouraging the widespread adoption of carbon pricing schemes. Economies can provide tax rebates proportional to production in emitting industries to support competitiveness, along with applying border carbon adjustments to tax carbon-intensive imports. Yet, they cannot avoid distortions and leakage unless carbon pricing schemes are extended to a broader set of countries worldwide.

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